Block practice · Integration, areas, volumes and kinematics

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Velocity and acceleration by differentiating

When is the particle at rest (or at a given velocity)?

Integrating trig functions using identities

Volume of revolution about the x-axis

Area between a curve and the x-axis

Area between two curves (or a curve and a line)

Volumes of revolution between two boundaries

Practice workbook · Velocity and acceleration by differentiating

Differentiating displacement or velocity with respect to time and substituting a time.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

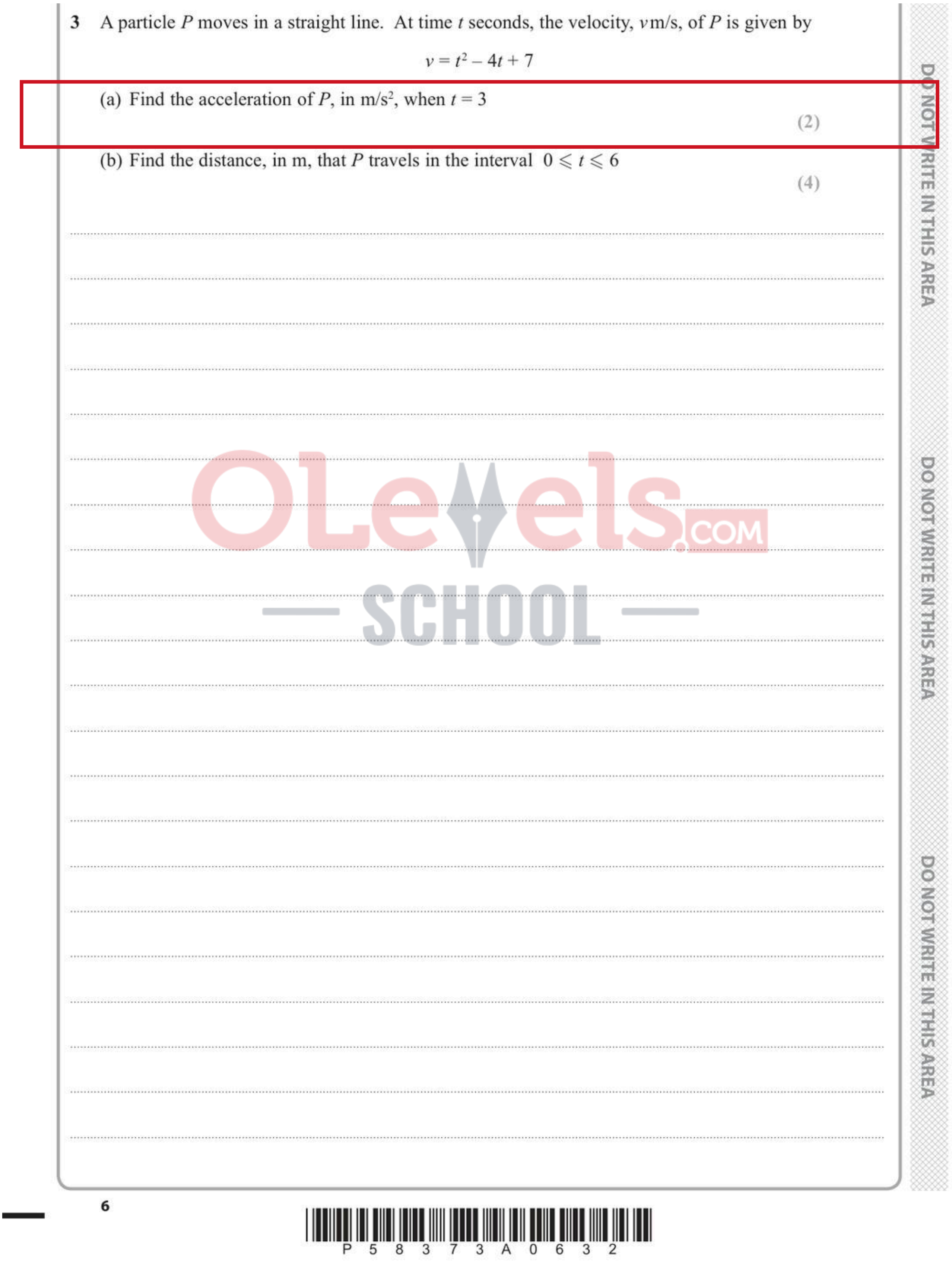
The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-25

1. Question 3 · part a · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a only.



3 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = t^2 - 4t + 7$$

(a) Find the acceleration of P, in m/s^2, when $t = 3$

(2)

(b) Find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 6$

(4)

1. Question 3 · part a · 2 marks · Working and answer

Working and answer

2. Question 1 · part b · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

1 A particle P moves in a straight line. At time t seconds, the velocity, v m/s, of P is given by

$$v = 3t^2 - 9t + 7$$

(a) Show that P never comes to rest.

(2)

(b) Find the acceleration of P, in m/s^2, when $t = 4$

(2)

Using algebra

(c) find the distance, in m, that P travels in the interval $0 \leqslant t \leqslant 5$

(3)

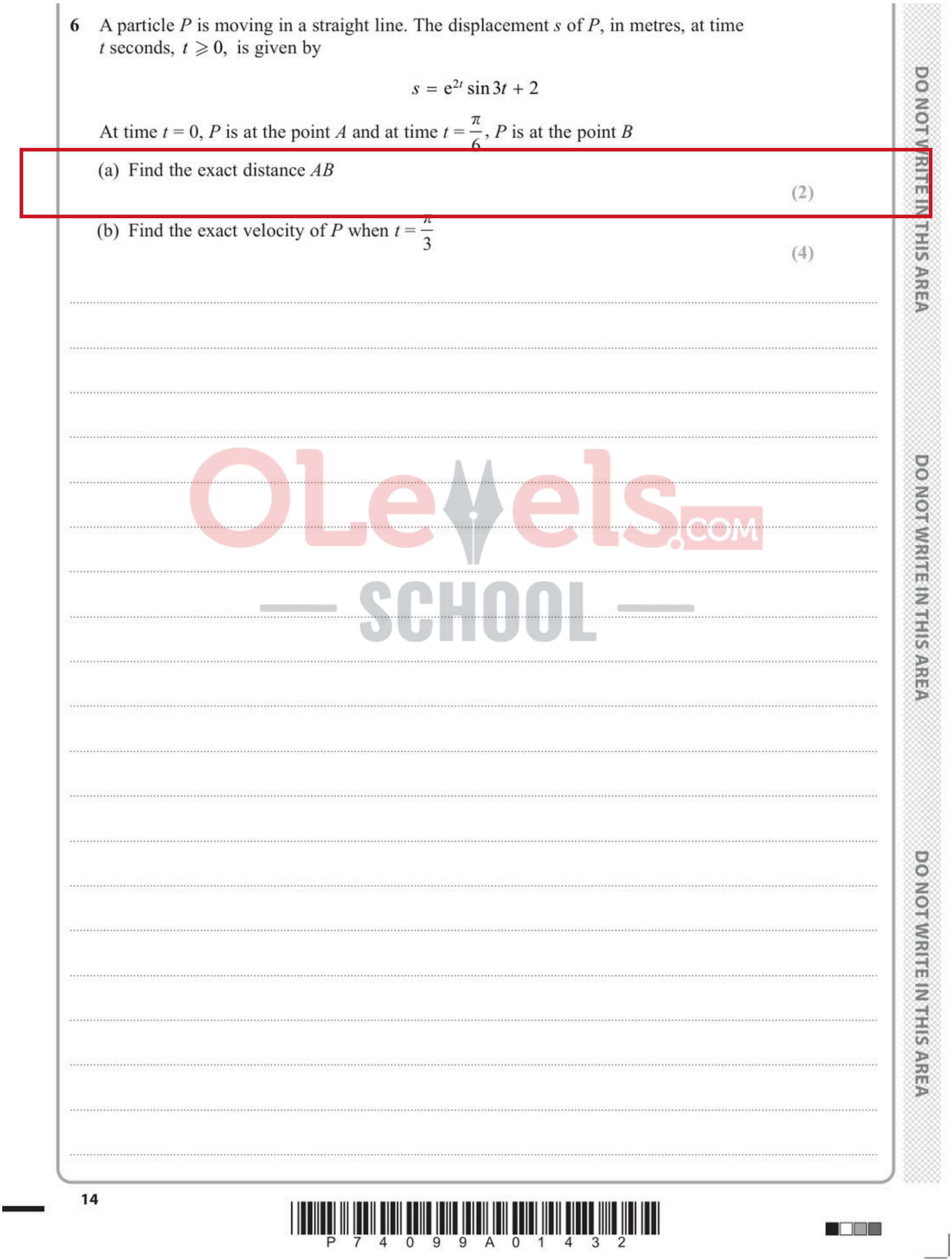
2. Question 1 · part b · 2 marks · Working and answer

Working and answer

3. Question 6 · part a · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.



6 A particle P is moving in a straight line. The displacement s of P, in metres, at time t seconds, $t \geqslant 0$, is given by

$$s = e^{2t}\sin 3t + 2$$

At time $t = 0$, P is at the point A and at time $t = \frac{\pi}{6}$, P is at the point B

(a) Find the exact distance AB

(2)

(b) Find the exact velocity of P when $t = \frac{\pi}{3}$

(4)

3. Question 6 · part a · 2 marks · Working and answer

Working and answer

4. Question 6 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 A particle P is moving in a straight line. The displacement s of P, in metres, at time t seconds, $t \geqslant 0$, is given by

$$s = e^{2t}\sin 3t + 2$$

At time $t = 0$, P is at the point A and at time $t = \frac{\pi}{6}$, P is at the point B

(a) Find the exact distance AB

(2)

(b) Find the exact velocity of P when $t = \frac{\pi}{3}$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4. Question 6 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 6 continued

(Total for Question 6 is 6 marks)

4. Question 6 · part b · 4 marks · Working and answer

Working and answer

5. Question 2 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 A particle P is moving along a straight line, which passes through the fixed point O.

At time t seconds ($t \geqslant 0$), the velocity, v m/s, of P is given by

$$v = t^2 - 3t + 4$$

At time t seconds the acceleration of P is a m/s^2

(a) Find an expression for a in terms of t

(2)

The displacement of P from O is 7 m when $t = 2$

(b) Find the exact displacement of P from O when $t = 4$

(5)

5. Question 2 · part a · 2 marks · Working and answer

Working and answer

6. Question 4 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

4 A particle P is moving along the x-axis. At time t seconds, $t \geqslant 0$, the velocity, $v\,\text{m/s}$, of P is given by $v = 2t^2 - 16t + 30$

(a) Find the acceleration, in m/s^2, of P when $t = 5$

(2)

P comes to instantaneous rest at the points M and N at times t_1 seconds and t_2 seconds where $t_2 > t_1$

(b) Find the exact distance MN

(8)

6. Question 4 · part a · 2 marks · Working and answer

Working and answer

7. Question 5 · part a · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

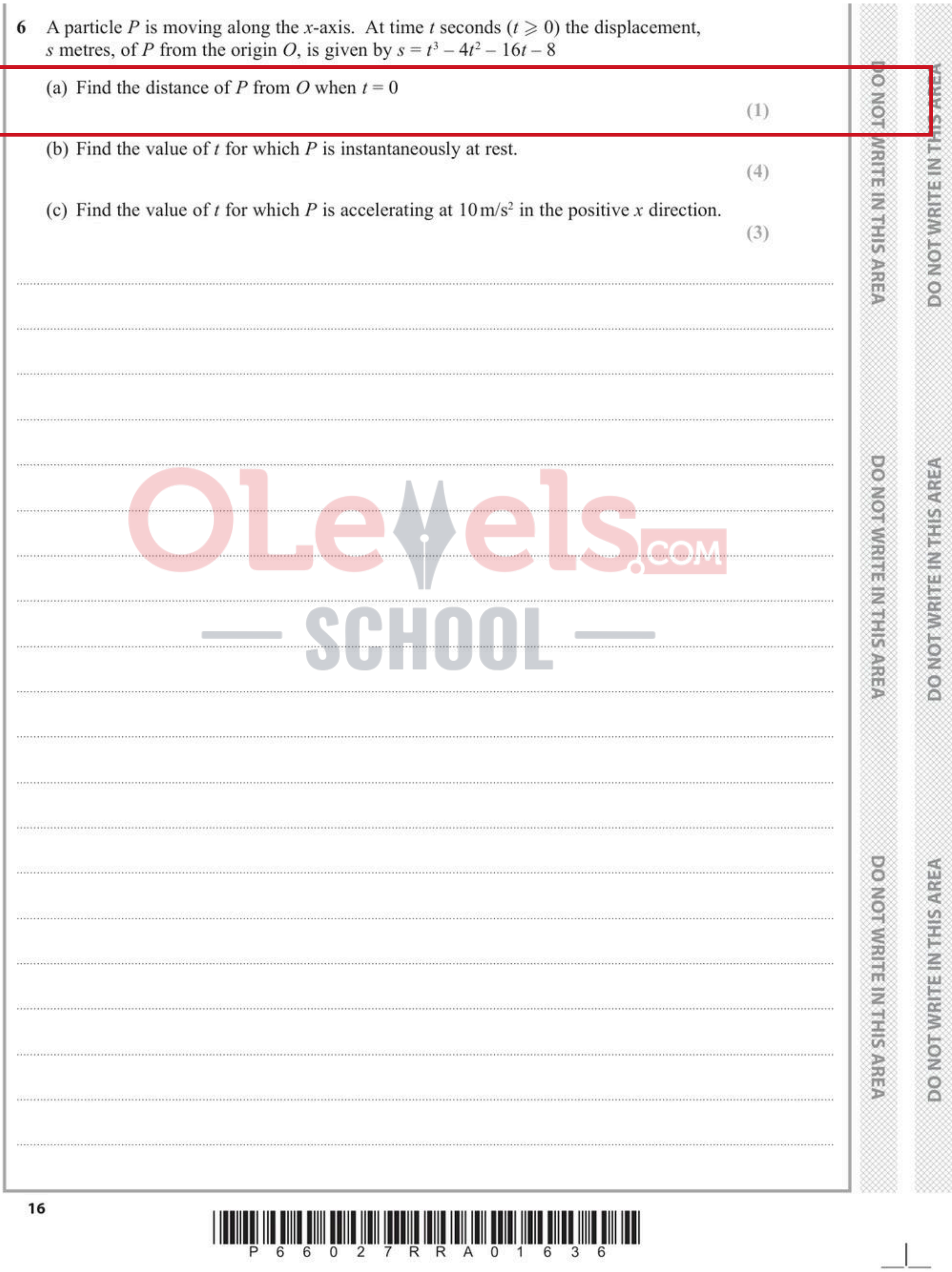
7. Question 5 · part a · 2 marks · Working and answer

Working and answer

8. Question 6 · part a · 1 mark

2021 NOV · Paper 02

Further practice · Attempt part a only.



6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

(c) Find the value of t for which P is accelerating at 10 m/s^2 in the positive x direction.

(3)

8. Question 6 · part a · 1 mark · Working and answer

Working and answer

9. Question 2 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 2 · part a · 2 marks · Working and answer

Working and answer

10. Question 2 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

10. Question 2 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 2 continued

(Total for Question 2 is 6 marks)

10. Question 2 · part c · 2 marks · Working and answer

Working and answer

11. Question 7 · part a · 1 mark

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

11. Question 7 · part a · 1 mark · Working and answer

Working and answer

12. Question 7 · part c · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 A particle P is moving along the x-axis. At time t seconds $(t \geqslant 0)$ the velocity of P is v m/s where

$$v = t^2 - 10t + 28$$

(a) Find the velocity of P when $t = 1$

(1)

Given that the distance of P from the origin is 24 m when $t = 3$

(b) find the distance of P from the origin when $t = 5$

(5)

(c) Find the acceleration of P when $t = 9$

(2)

(d) (i) Show that there are no values of t for which P is instantaneously at rest.

(ii) Find the least magnitude of the velocity of P

(3)

12. Question 7 · part c · 2 marks · Working and answer

Working and answer

Practice workbook · When is the particle at rest (or at a given velocity)?

Setting velocity (or acceleration or displacement) equal to a value and solving for t.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-26

1. Question 5 · part a · 3 marks

2022 JAN · Paper 02

Worked-example retry · Attempt part a only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

1. Question 5 · part a · 3 marks · Working and answer

Working and answer

2. Question 3 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

3 A particle P moves along the x-axis.

At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 6t - 16$

When $t = 0$, P is at the origin and is moving with velocity 12 m/s.

(a) Find an expression in terms of t for

(i) the velocity of P at time t seconds

(ii) the displacement of P at time t seconds.

(4)

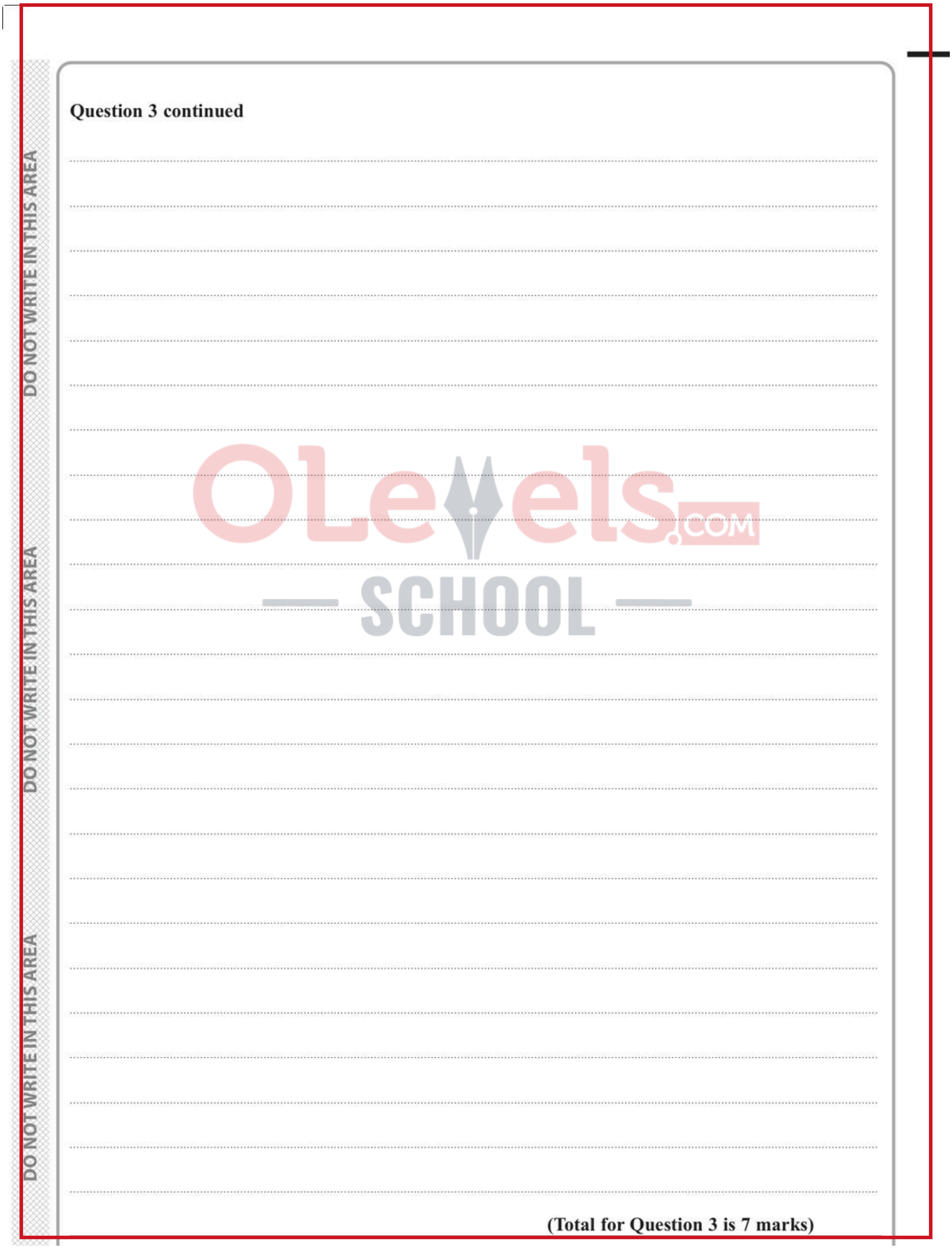
(b) Hence find the time at which P first returns to the origin.

(3)

2. Question 3 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

(Total for Question 3 is 7 marks)

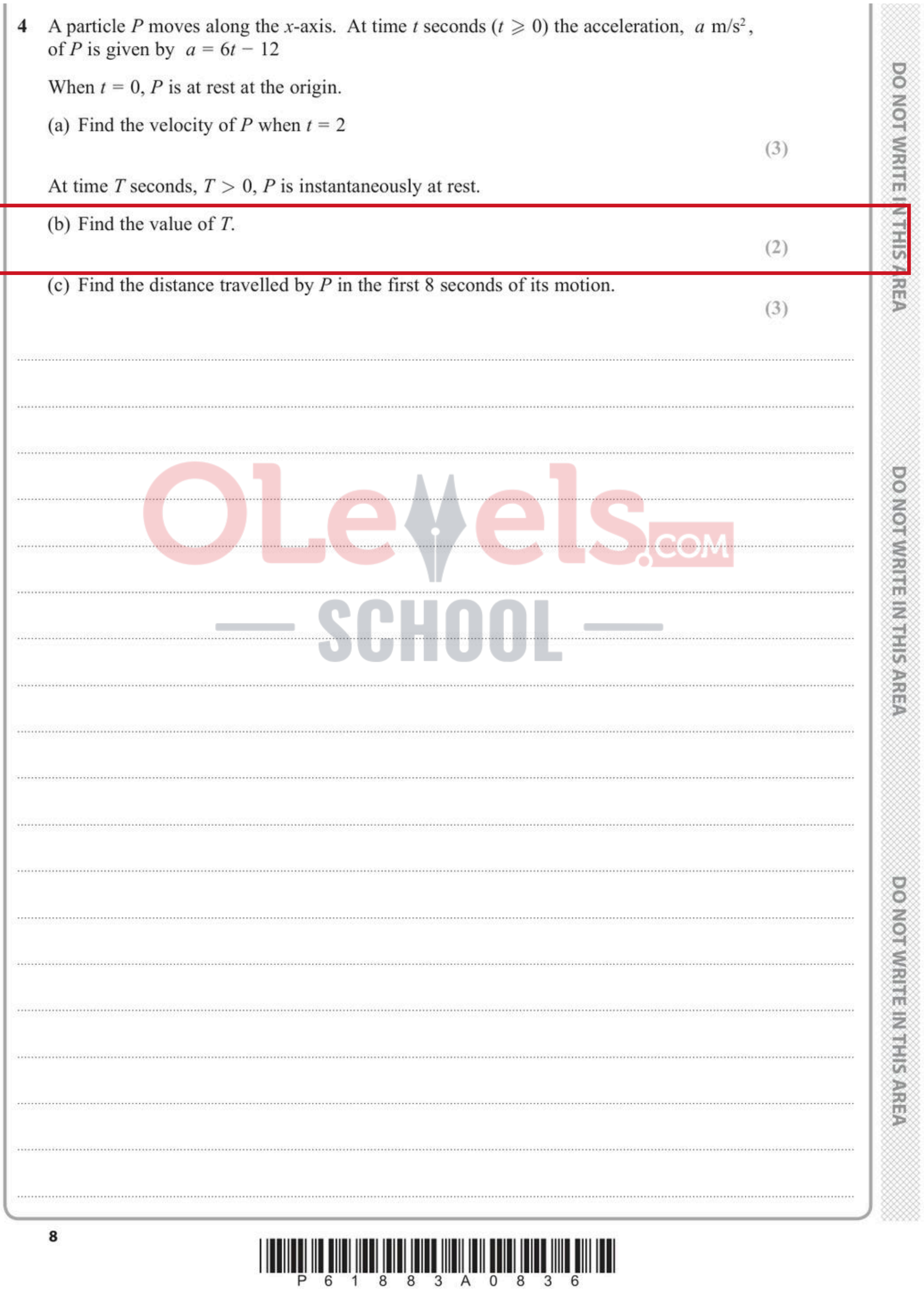
2. Question 3 · part b · 3 marks · Working and answer

Working and answer

3. Question 4 · part b · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



4 A particle P moves along the x-axis. At time t seconds ($t \geqslant 0$) the acceleration, a m/s^2, of P is given by $a = 6t - 12$

When $t = 0$, P is at rest at the origin.

(a) Find the velocity of P when $t = 2$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T.

(2)

(c) Find the distance travelled by P in the first 8 seconds of its motion.

(3)

8

P 6 1 8 8 3 A 0 8 3 6

3. Question 4 · part b · 2 marks · Working and answer

Working and answer

4. Question 1 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

1 A particle P is moving in a straight line. At time t seconds, $t \geqslant 0$, the displacement, s metres, of P from a fixed point O of the line is given by

$$s = 3 + 8t + t^2 - \frac{1}{3}t^3$$

Find the distance of P from O when P is instantaneously at rest.

(4)

(Total for Question 1 is 4 marks)

4. Question 1 · part whole · 4 marks · Working and answer

Working and answer

5. Question 5 · part b · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

5 A particle P is moving along the x-axis.
At time t seconds $(t \geqslant 0)$ the acceleration, a m/s^2, of P is given by $a = 3t - 4$

When $t = 0$, P is at rest.

(a) Find the velocity of P when $t = 4$

(3)

At time T seconds, $T > 0$, P is instantaneously at rest.

(b) Find the value of T

(2)

When $t = 0$, P is at the point with coordinates $(-10, 0)$

(c) Find the displacement of P from the origin when $t = 3$

(4)

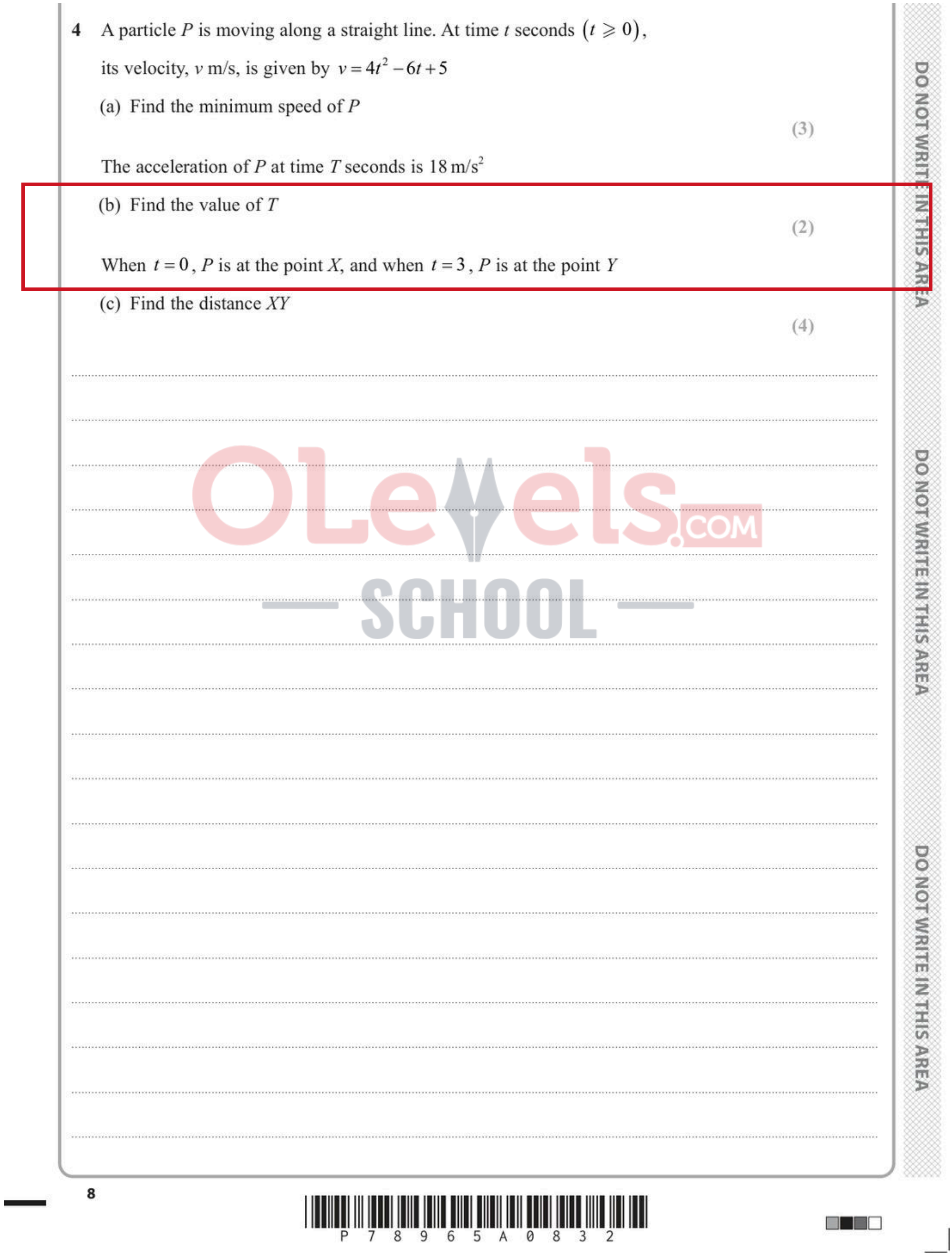
5. Question 5 · part b · 2 marks · Working and answer

Working and answer

6. Question 4 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.



4 A particle P is moving along a straight line. At time t seconds $(t \geqslant 0)$, its velocity, v m/s, is given by $v = 4t^2 - 6t + 5$

(a) Find the minimum speed of P

(3)

The acceleration of P at time T seconds is 18 m/s^2

(b) Find the value of T

(2)

When $t = 0$, P is at the point X, and when $t = 3$, P is at the point Y

(c) Find the distance XY

(4)

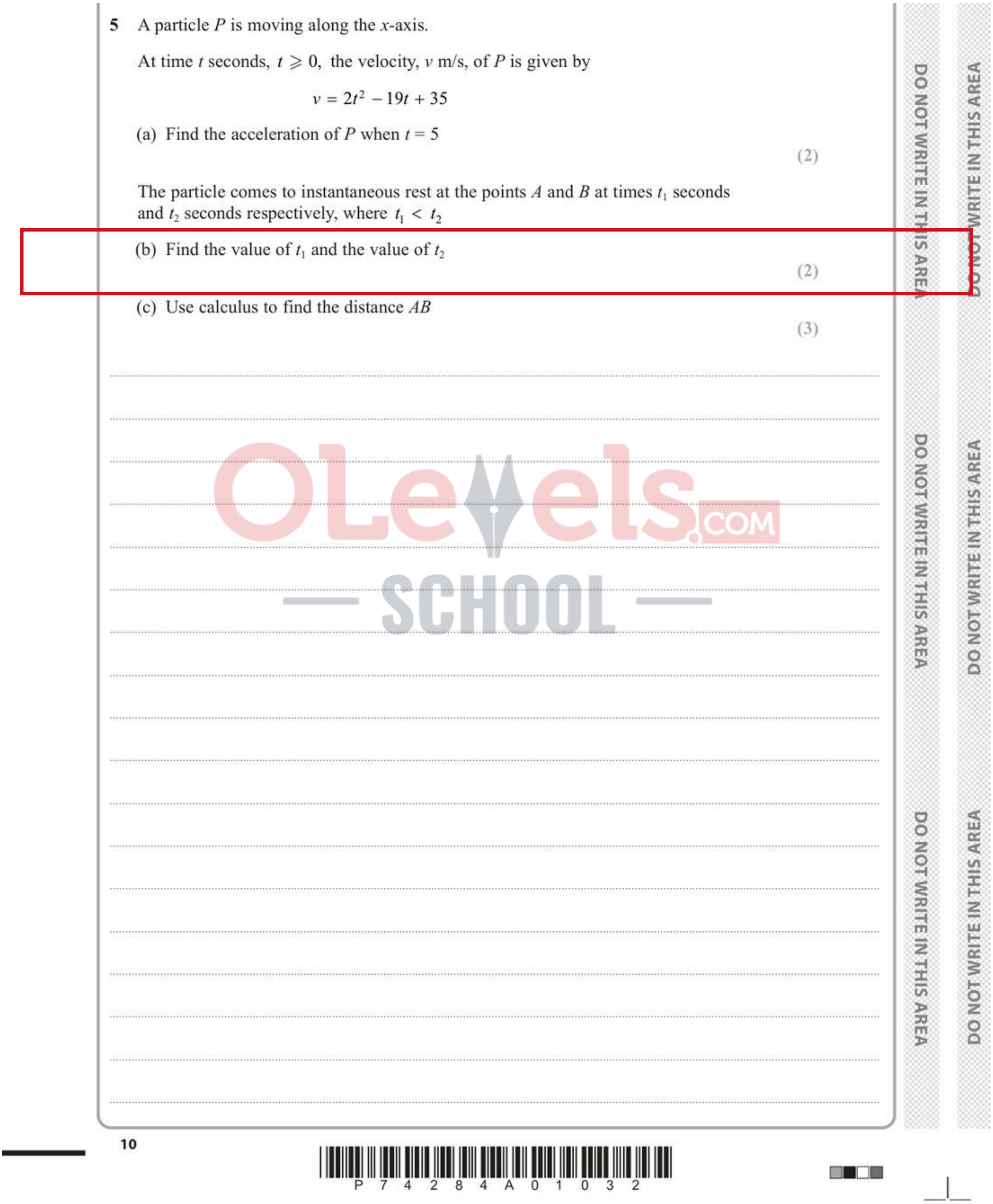
6. Question 4 · part b · 2 marks · Working and answer

Working and answer

7. Question 5 · part b · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



5 A particle P is moving along the x-axis.

At time t seconds, $t \geqslant 0$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

(a) Find the acceleration of P when $t = 5$

(2)

The particle comes to instantaneous rest at the points A and B at times t_1 seconds and t_2 seconds respectively, where $t_1 < t_2$

(b) Find the value of t_1 and the value of t_2

(2)

(c) Use calculus to find the distance AB

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

7. Question 5 · part b · 2 marks · Working and answer

Working and answer

8. Question 5 · part a · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

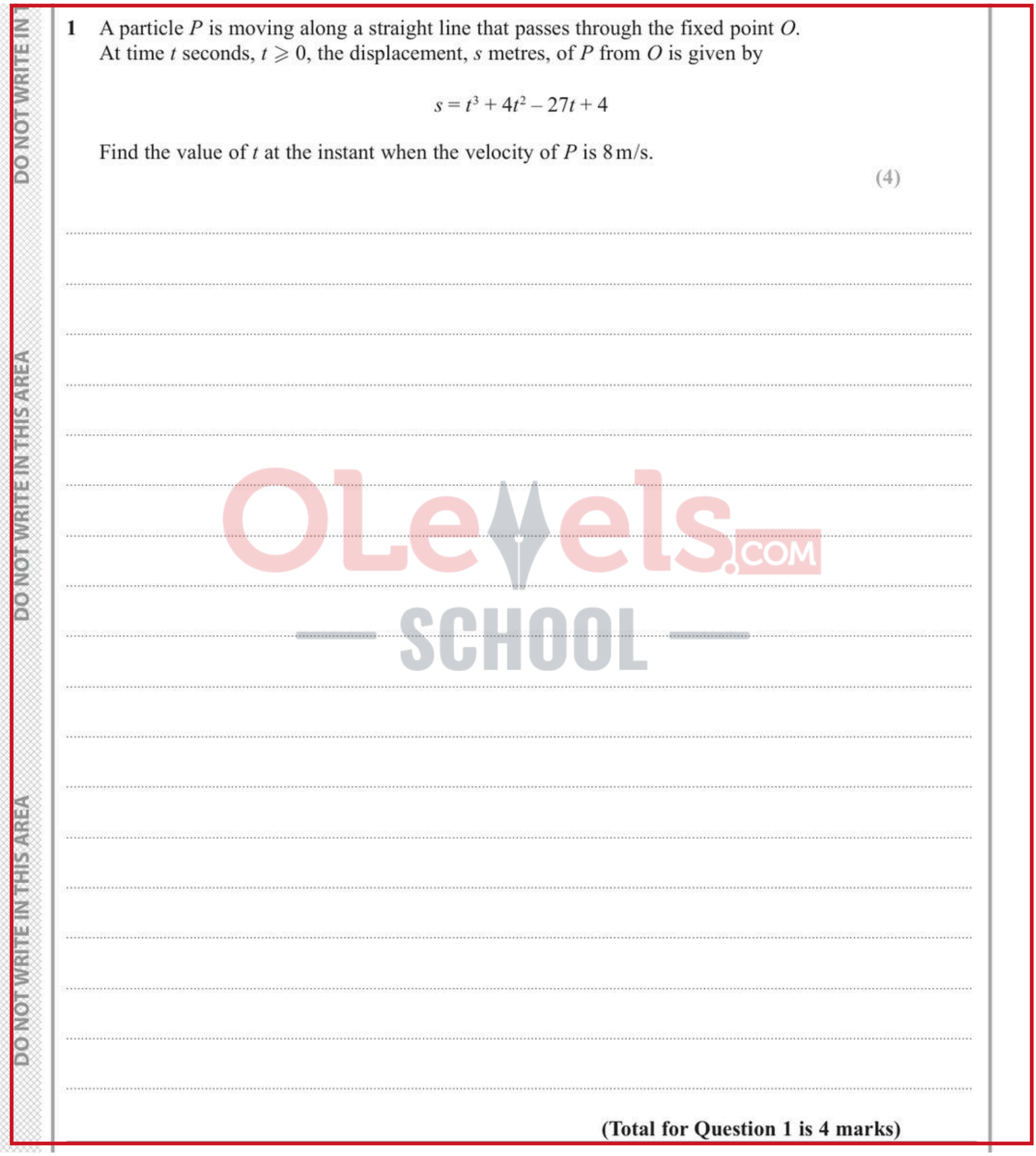
8. Question 5 · part a · 3 marks · Working and answer

Working and answer

9. Question 1 · part whole · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part whole only.



1 A particle P is moving along a straight line that passes through the fixed point O.
At time t seconds, $t \geqslant 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 4t^2 - 27t + 4$$

Find the value of t at the instant when the velocity of P is 8 m/s.

(4)

(Total for Question 1 is 4 marks)

9. Question 1 · part whole · 4 marks · Working and answer

Working and answer

10. Question 6 · part b · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

(c) Find the value of t for which P is accelerating at $10\,\text{m/s}^2$ in the positive x direction.

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

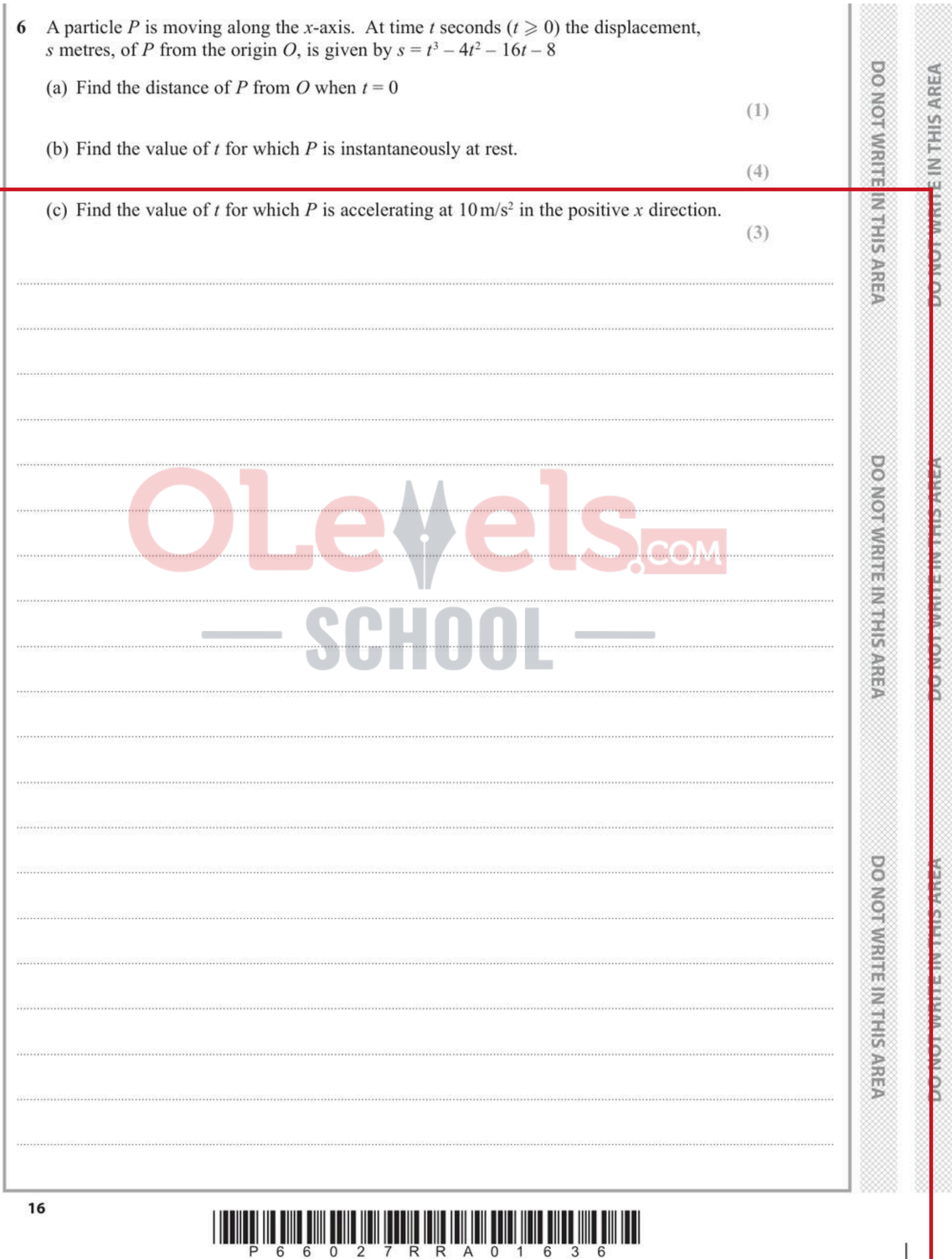
10. Question 6 · part b · 4 marks · Working and answer

Working and answer

11. Question 6 · part c · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.



6 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the displacement, s metres, of P from the origin O, is given by $s = t^3 - 4t^2 - 16t - 8$

(a) Find the distance of P from O when $t = 0$

(1)

(b) Find the value of t for which P is instantaneously at rest.

(4)

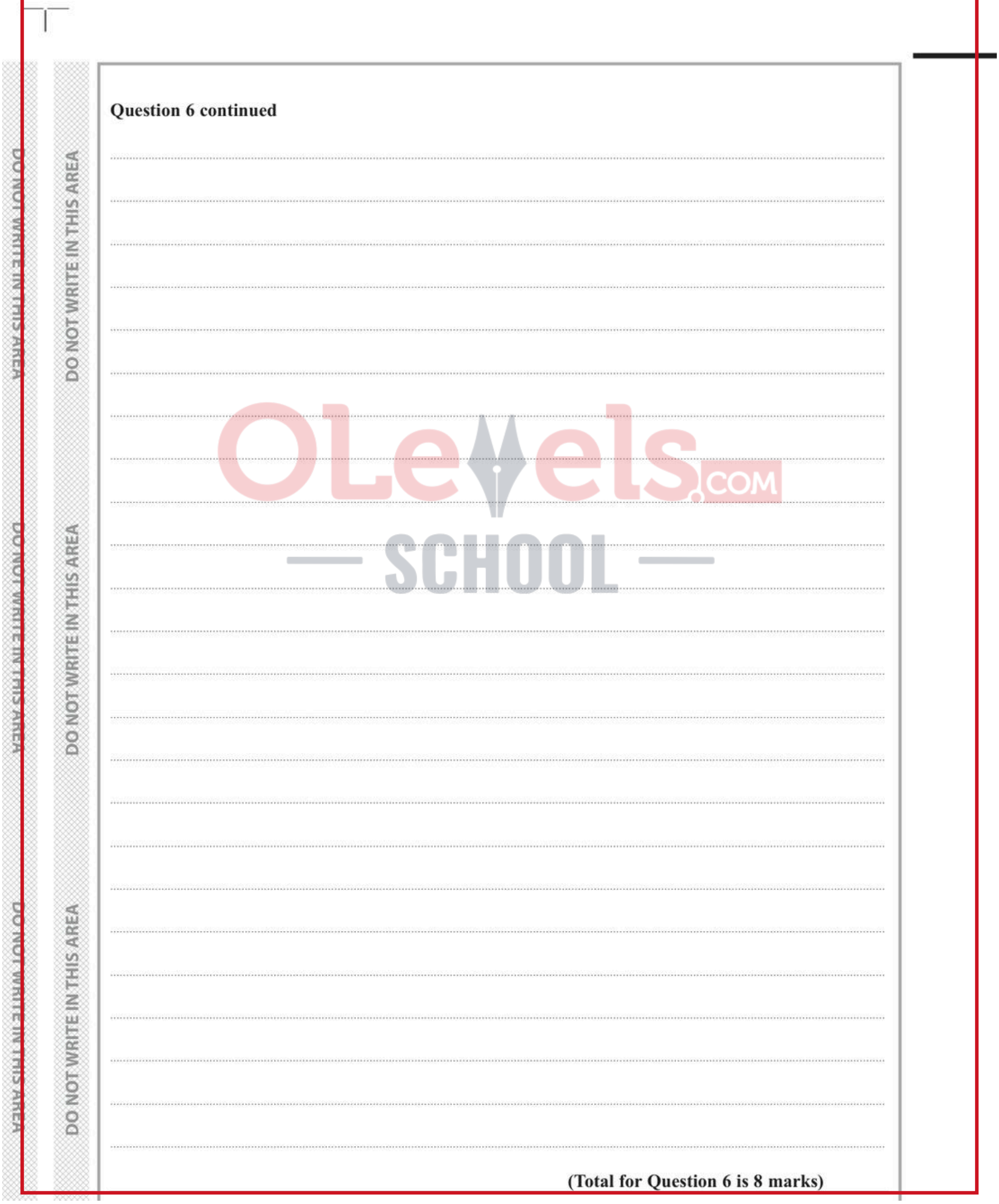
(c) Find the value of t for which P is accelerating at $10\,\text{m/s}^2$ in the positive x direction.

(3)

11. Question 6 · part c · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 6 continued

(Total for Question 6 is 8 marks)

11. Question 6 · part c · 3 marks · Working and answer

Working and answer

12. Question 2 · part b · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

2 A particle P moves along the x-axis. At time t seconds, the displacement, x metres, of P from the origin O is given by

$$x = t^4 - 13.5t + 12$$

(a) Find the velocity, in m/s, of P when $t = 3$

(2)

(b) Find the value of t for which P is instantaneously at rest.

(2)

(c) Find the acceleration, in m/s^2, of P when $t = 2$

(2)

12. Question 2 · part b · 2 marks · Working and answer

Working and answer

13. Question 3 · part a · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

13. Question 3 · part a · 3 marks · Working and answer

Working and answer

14. Question 3 · part b · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

3 A particle P is moving along a straight line, from a fixed origin O

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 2t^2 - 19t + 35$$

At time t seconds the acceleration of P is a m/s^2

(a) Find the value of t for which $a = 0$

(3)

P is instantaneously at rest at time T_1 seconds and at time T_2 seconds where $T_2 > T_1$

(b) Find the value of T_1 and the value of T_2

(3)

(c) Find the exact distance, in metres, that P travels between the times T_1 and T_2
Show your working clearly.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

14. Question 3 · part b · 3 marks · Working and answer

Working and answer

15. Question 5 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

5 A particle P is moving along the x-axis. At time t seconds ($t \geqslant 0$) the velocity, v m/s, of P is given by $v = 3t^2 - 23t + 30$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

When $t = 0$, P is at the point with coordinates $(d, 0)$

Given that, when $t = 8$, P is at the point with coordinates $(26, 0)$

(c) find the value of d

(4)

15. Question 5 · part b · 2 marks · Working and answer

Working and answer

16. Question 5 · part b · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

5 A particle P is moving along a straight line.

At time t seconds $(t \geqslant 0)$, the velocity, v m/s, of P is given by

$$v = 3t^2 - 16t + 5$$

(a) Find the values of t when P is instantaneously at rest.

(3)

At time t seconds the acceleration of P is a m/s^2

(b) Find the range of values of t for which $a > 0$

(2)

(c) Find the distance, in m, that P travels in the interval $1 \leqslant t \leqslant 4$

(5)

16. Question 5 · part b · 2 marks · Working and answer

Working and answer

Practice workbook · Integrating trig functions using identities

Rewriting trig products or powers as sums of sin(ax) and cos(ax) with identities, then integrating and evaluating exactly.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-17

1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

6 (a) Show that

$$\sin(A + B) + \sin(A - B) = 2\sin A \cos B$$

(2)

(b) Hence express $2\sin 7x \cos x$ in the form $\sin mx + \sin nx$ where m and n are integers, giving the value of m and the value of n.

(1)

(c) Use calculus to evaluate

$$\int_0^{\frac{\pi}{4}} (6\sin 7x \cos x)\, dx$$

(4)

1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

Question 6 continued

1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

Question 6 continued

1. Question 6 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part c only.

Question 6 continued

(Total for Question 6 is 7 marks)

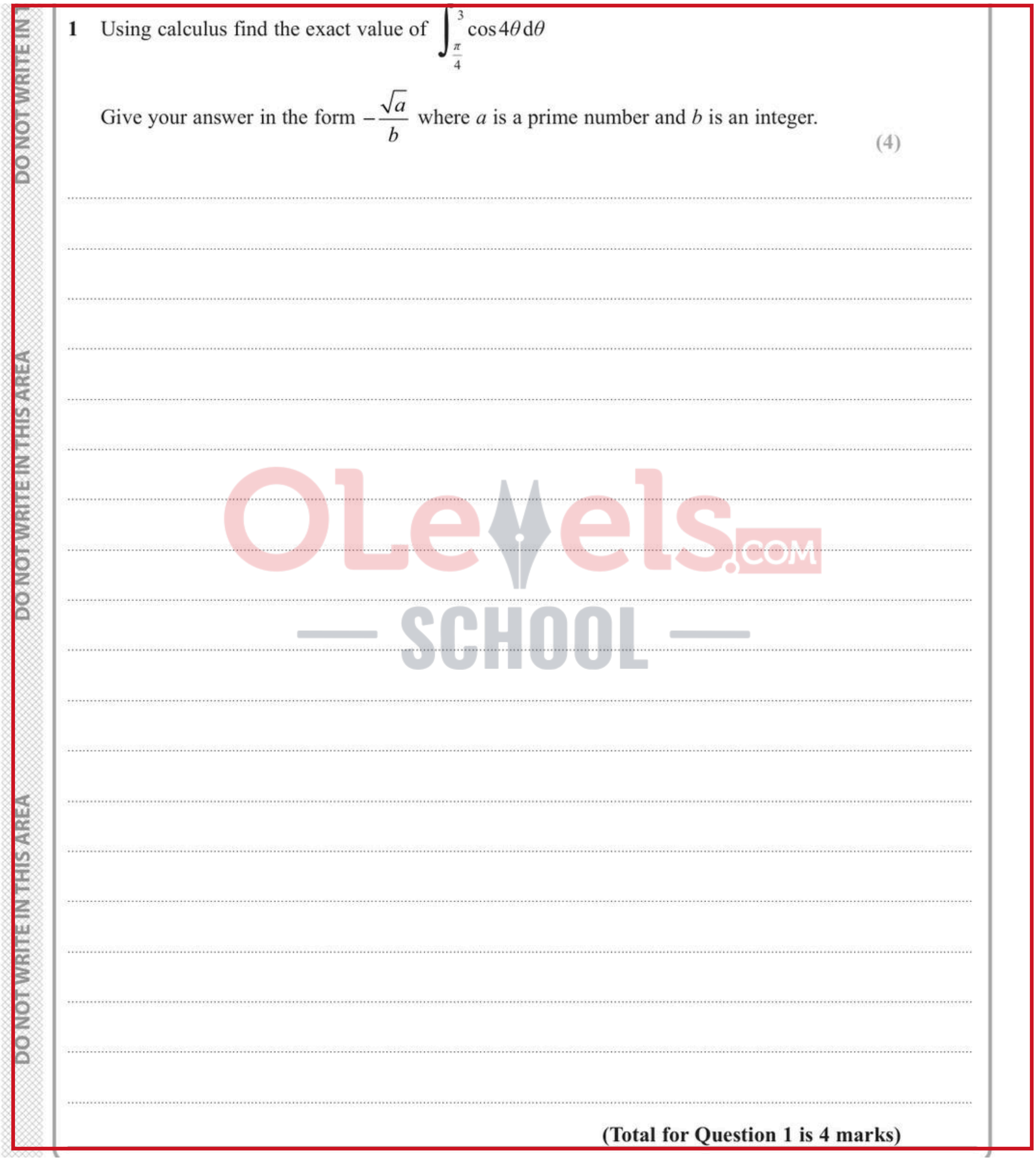
1. Question 6 · part c · 4 marks · Working and answer

Working and answer

2. Question 1 · part whole · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part whole only.



1 Using calculus find the exact value of $\int_{\frac{\pi}{4}}^{3} \cos 4\theta \, d\theta$

Give your answer in the form $-\frac{\sqrt{a}}{b}$ where a is a prime number and b is an integer.

(4)

(Total for Question 1 is 4 marks)

2. Question 1 · part whole · 4 marks · Working and answer

Working and answer

3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10 (a) Use the formula for $\cos(A+B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4}+\frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4}+\frac{\pi}{24}\right)-1\right] = -\frac{1}{16}$ $\quad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

Turn over ▶

3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

3. Question 10 · part d · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 10 continued

(Total for Question 10 is 15 marks)

3. Question 10 · part d · 4 marks · Working and answer

Working and answer

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

10 (a) Show that

$$\cos(A+B)+\cos(A-B)=2\cos A\cos B$$

(2)

(b) Hence show that

$$\cos P+\cos Q = 2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x\cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x\cos^2 x \,\mathrm{d}x$$

(4)

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

Question 10 continued

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

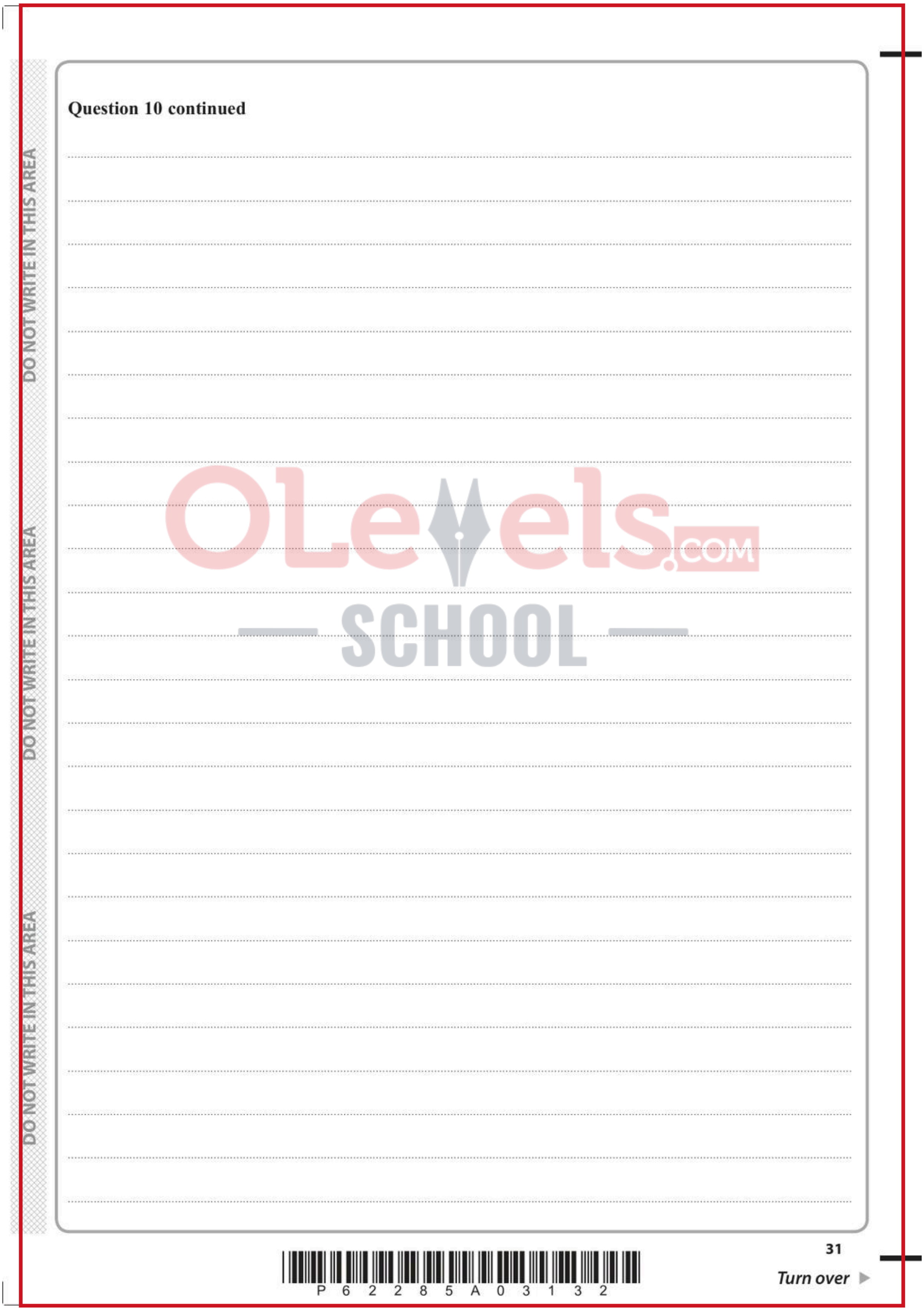
Further practice · Attempt part e only.

Question 10 continued

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.



Question 10 continued

4. Question 10 · part e · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part e only.

Question 10 continued

(Total for Question 10 is 16 marks)

4. Question 10 · part e · 4 marks · Working and answer

Working and answer

5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 11 continued

Turn over

5. Question 11 · part c · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 11 continued

(Total for Question 11 is 15 marks)

5. Question 11 · part c · 4 marks · Working and answer

Working and answer

6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$ **(3)**

(ii) $\sin 2A = 2\sin A\cos A$ **(1)**

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ **(4)**

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

6. Question 11 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

(Total for Question 11 is 16 marks)

6. Question 11 · part d · 4 marks · Working and answer

Working and answer

7. Question 9 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

Diagram **NOT** accurately drawn

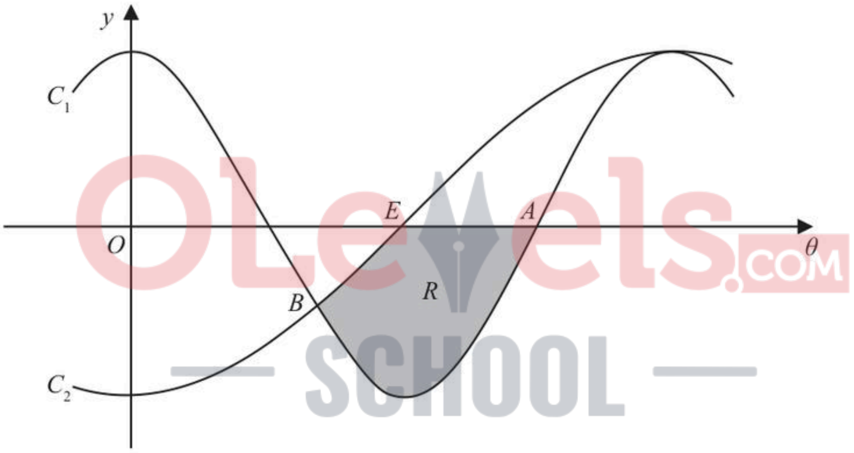


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

7. Question 9 · part b · 4 marks · Working and answer

Working and answer

8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 Using formulae from page 2

(a) show that $\cos(A-B) - \cos(A+B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

8. Question 10 · part d · 6 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

(Total for Question 10 is 16 marks)

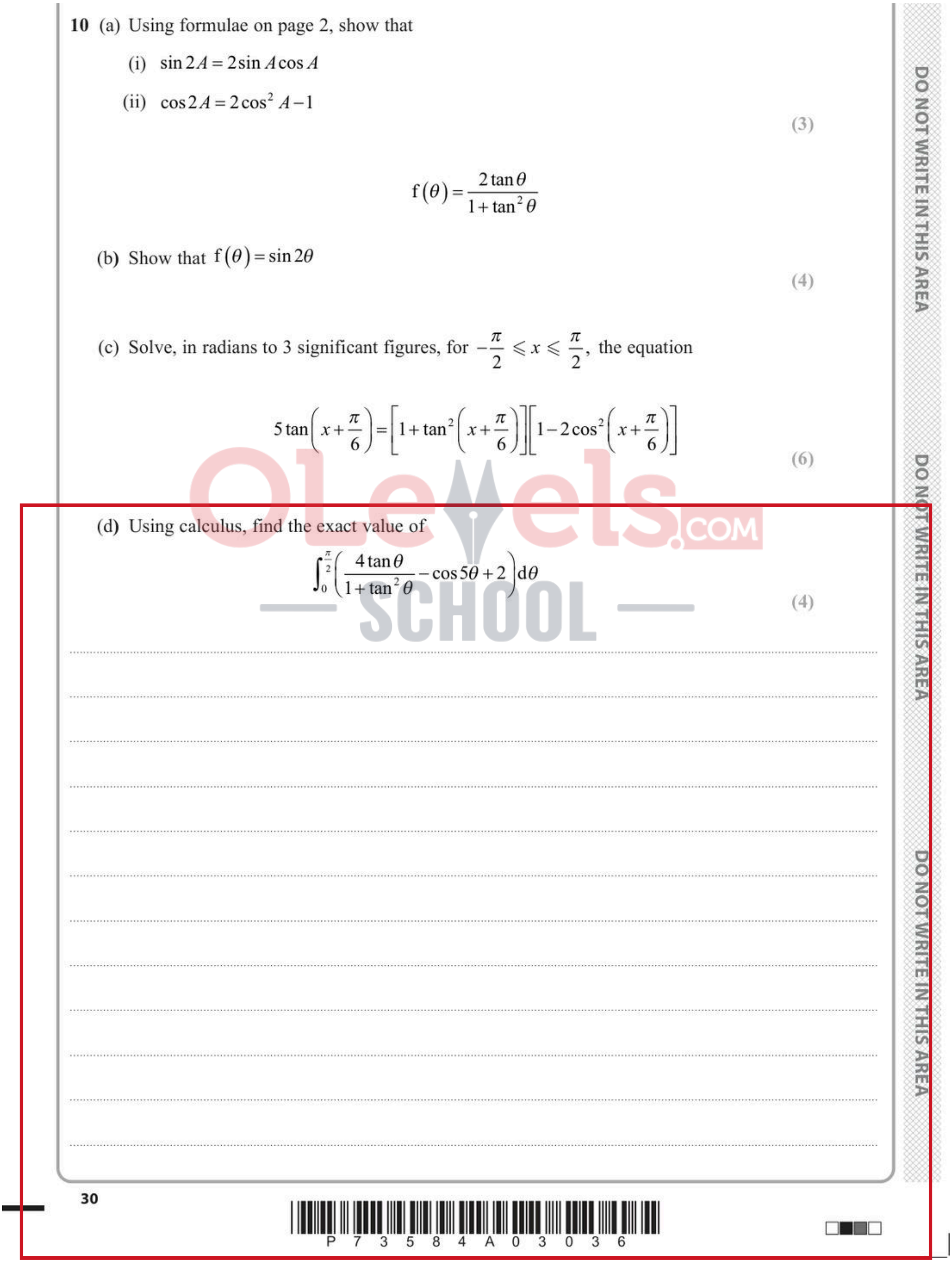
8. Question 10 · part d · 6 marks · Working and answer

Working and answer

9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.



10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}}\left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right)d\theta$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

30

P 7 3 5 8 4 A 0 3 0 3 6

9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

Question 10 continued

9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

Question 10 continued

9. Question 10 · part d · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part d only.

Question 10 continued

(Total for Question 10 is 17 marks)

9. Question 10 · part d · 4 marks · Working and answer

Working and answer

Practice workbook · Volume of revolution about the x-axis

Squaring y and finding pi times the integral of y^2 dx between limits, including solving for an unknown limit.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-22

1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

6 (a) Using a formula from page 2, show that $\cos 2A = 1 - 2\sin^2 A$

(2)

The finite region R is bounded by the curve with equation $y = 3 + 2\sin x$, the x-axis, the y-axis and the line with equation $x = \frac{\pi}{4}$

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.
Give your answer to the nearest integer.

(6)

1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

Question 6 continued

(Total for Question 6 is 8 marks)

1. Question 6 · part b · 6 marks · Working and answer

Working and answer

2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

11 Given that

$$y = \frac{e^{4x}}{32}(8x^2 - 4x + 1)$$

(a) show that $\frac{dy}{dx} = x^2e^{4x}$

(5)

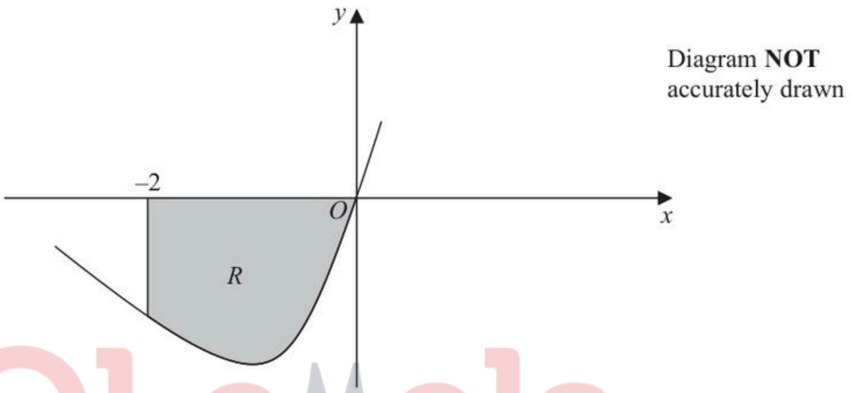


Figure 2

Figure 2 shows part of the curve C with equation $y = 3xe^{2x}$

The finite region R bounded by C, the straight line with equation $x = -2$ and the x-axis, shown shaded in Figure 2, is rotated though 360° about the x-axis.

(b) Using part (a), find the volume, to 2 significant figures, of the solid formed.

(4)

2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 11 continued

Turn over ▶

2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 11 continued

(Total for Question 11 is 9 marks)

2. Question 11 · part b · 4 marks · Working and answer

Working and answer

3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

11

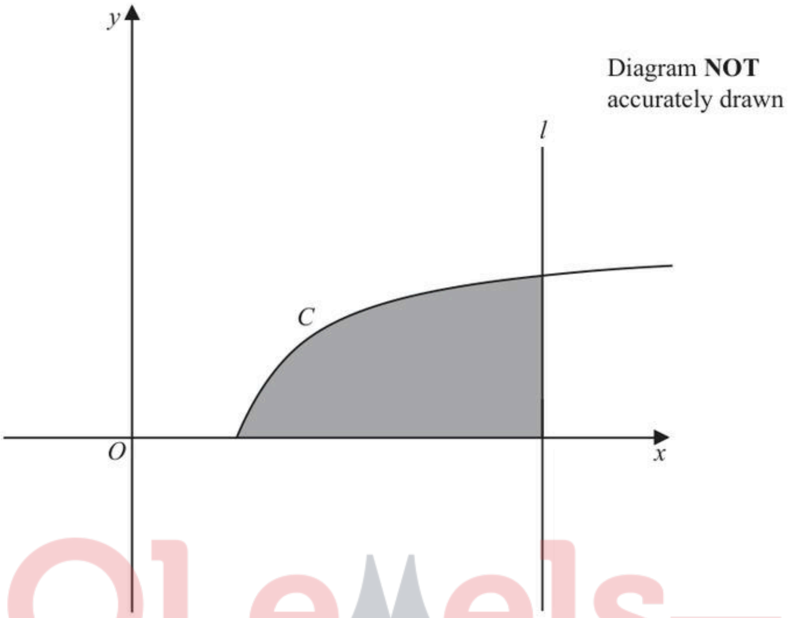


Figure 5

Figure 5 shows part of the curve C with equation $y = \sqrt{4x - 8}$ and the line l with equation $x = b$ where $b > 0$

The finite region bounded by the curve C, the x-axis and the line l, shown shaded in Figure 5, is rotated through $360°$ about the x-axis.

Given that the volume of the solid formed is 50π units^3

find the value of b

(7)

3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

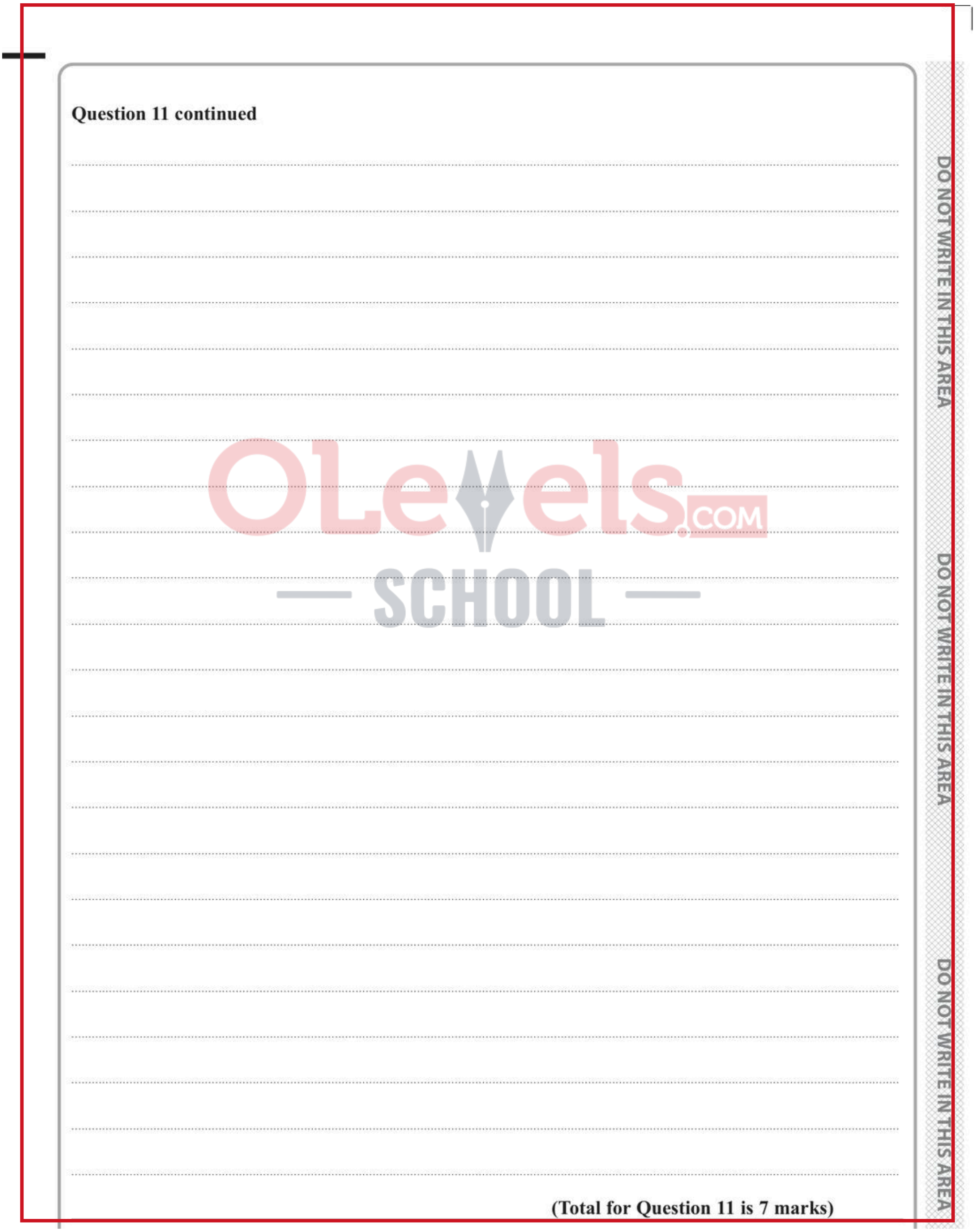
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 11 continued

3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.



Question 11 continued

(Total for Question 11 is 7 marks)

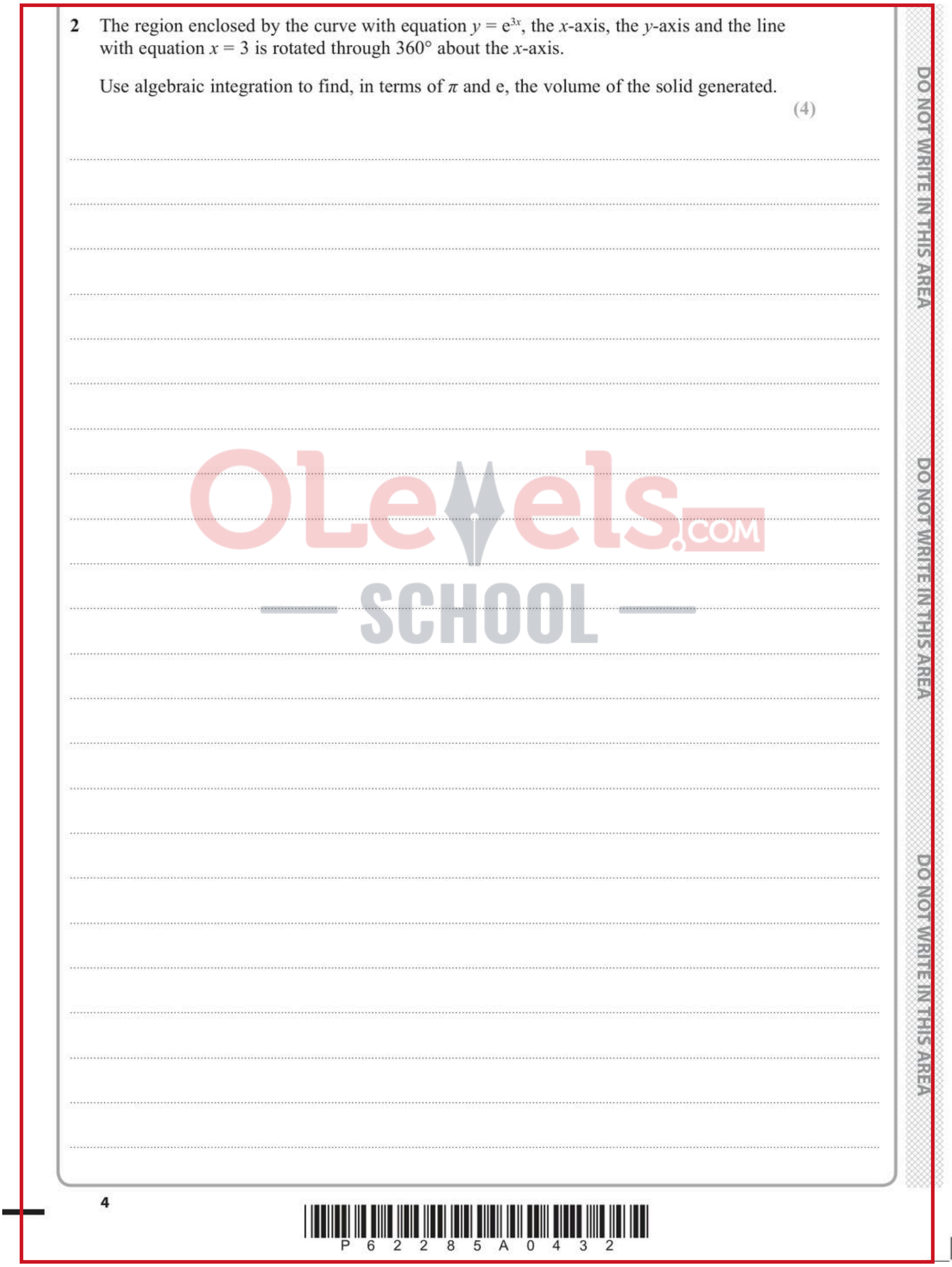
3. Question 11 · part whole · 7 marks · Working and answer

Working and answer

4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.



2 The region enclosed by the curve with equation $y = e^{3x}$, the x-axis, the y-axis and the line with equation $x = 3$ is rotated through 360° about the x-axis.

Use algebraic integration to find, in terms of π and e, the volume of the solid generated.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
4
P 6 2 2 8 5 A 0 4 3 2

4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

(Total for Question 2 is 4 marks)

4. Question 2 · part whole · 4 marks · Working and answer

Working and answer

5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7

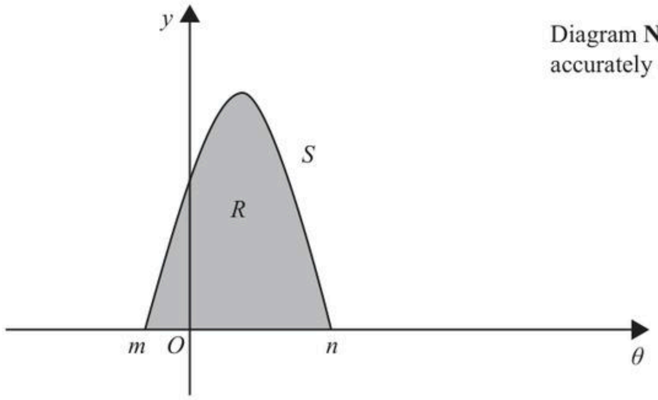


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve S with equation $y = \left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}}$

where $m \leqslant \theta \leqslant n$

The curve S meets the x-axis at the point with coordinates $(m, 0)$ and at the point with coordinates $(n, 0)$

(a) Find the exact value of m and the exact value of n

(3)

The finite region R, shown shaded in Figure 2, is bounded by the curve S, and the x-axis in the region $m \leqslant \theta \leqslant n$

The region R is rotated through 2π radians about the theta-axis.

(b) Use calculus to find the exact volume of the solid generated.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

Turn over ▶

5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

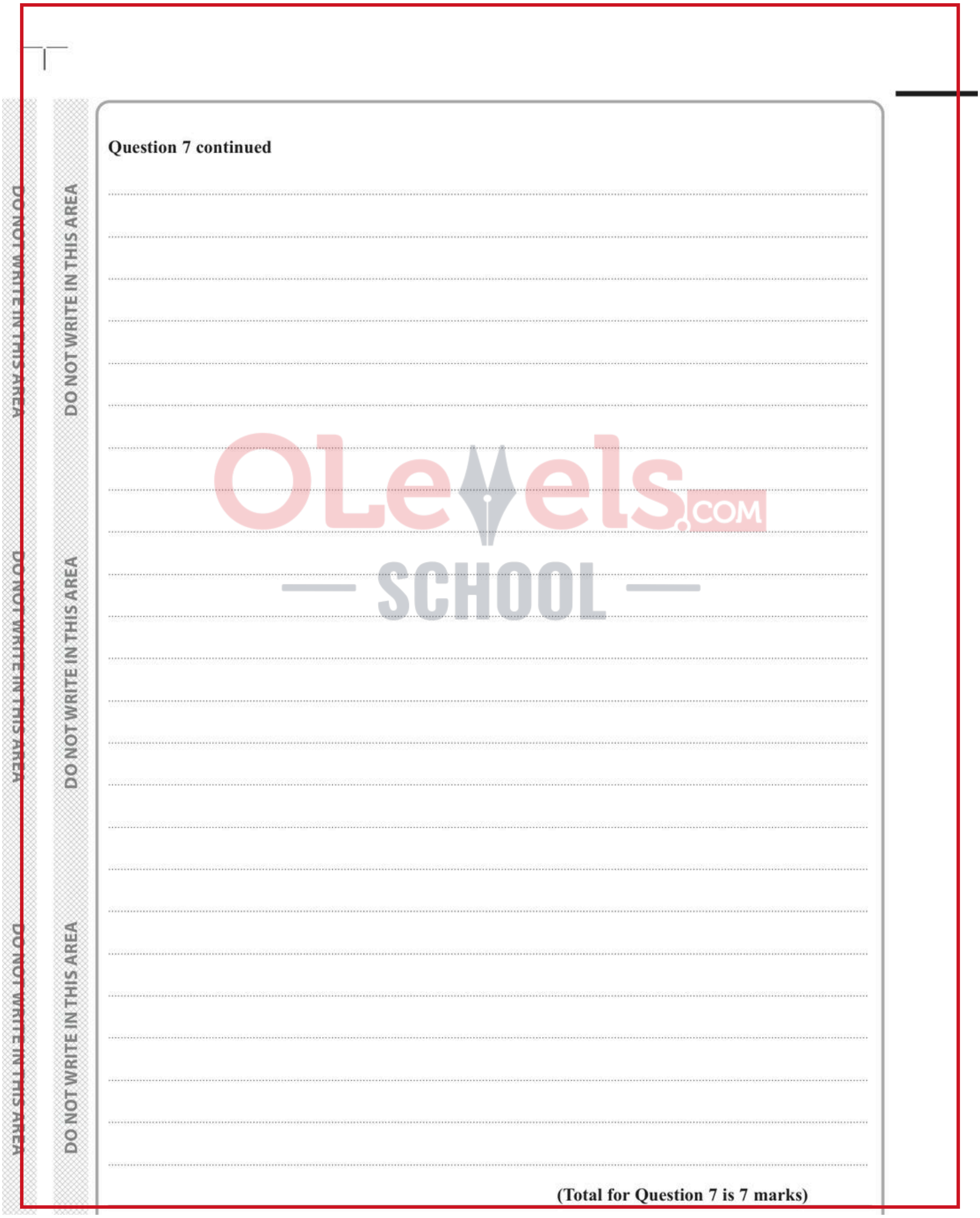
Question 7 continued

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

Question 7 continued

(Total for Question 7 is 7 marks)

5. Question 7 · part b · 4 marks · Working and answer

Working and answer

6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

11

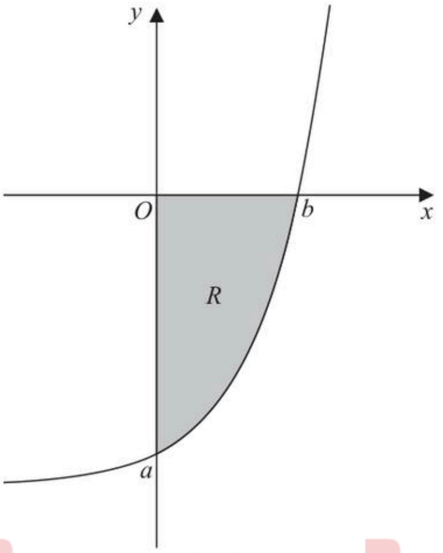


Diagram **NOT** accurately drawn

Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p\ln 3 + q)$, where p and q are integers.

(6)

6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

Turn over ▶

6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 11 continued

6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

(Total for Question 11 is 9 marks)

6. Question 11 · part b · 6 marks · Working and answer

Working and answer

Practice workbook · Area between a curve and the x-axis

Finding areas under one curve, splitting where it crosses the x-axis and adding the sizes of the separate pieces.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-19

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.

10

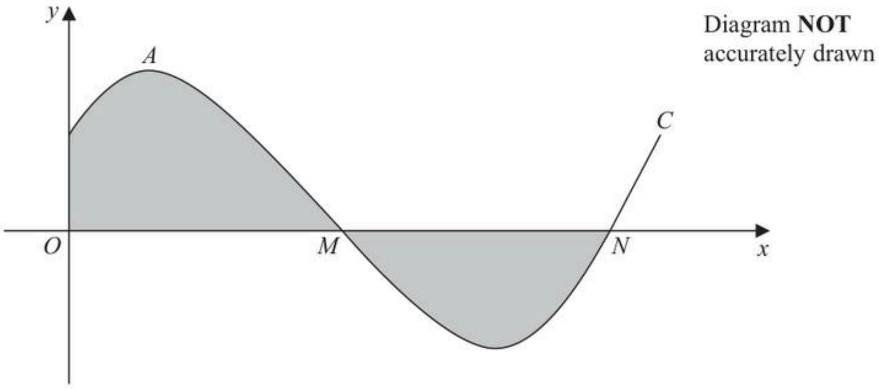


Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

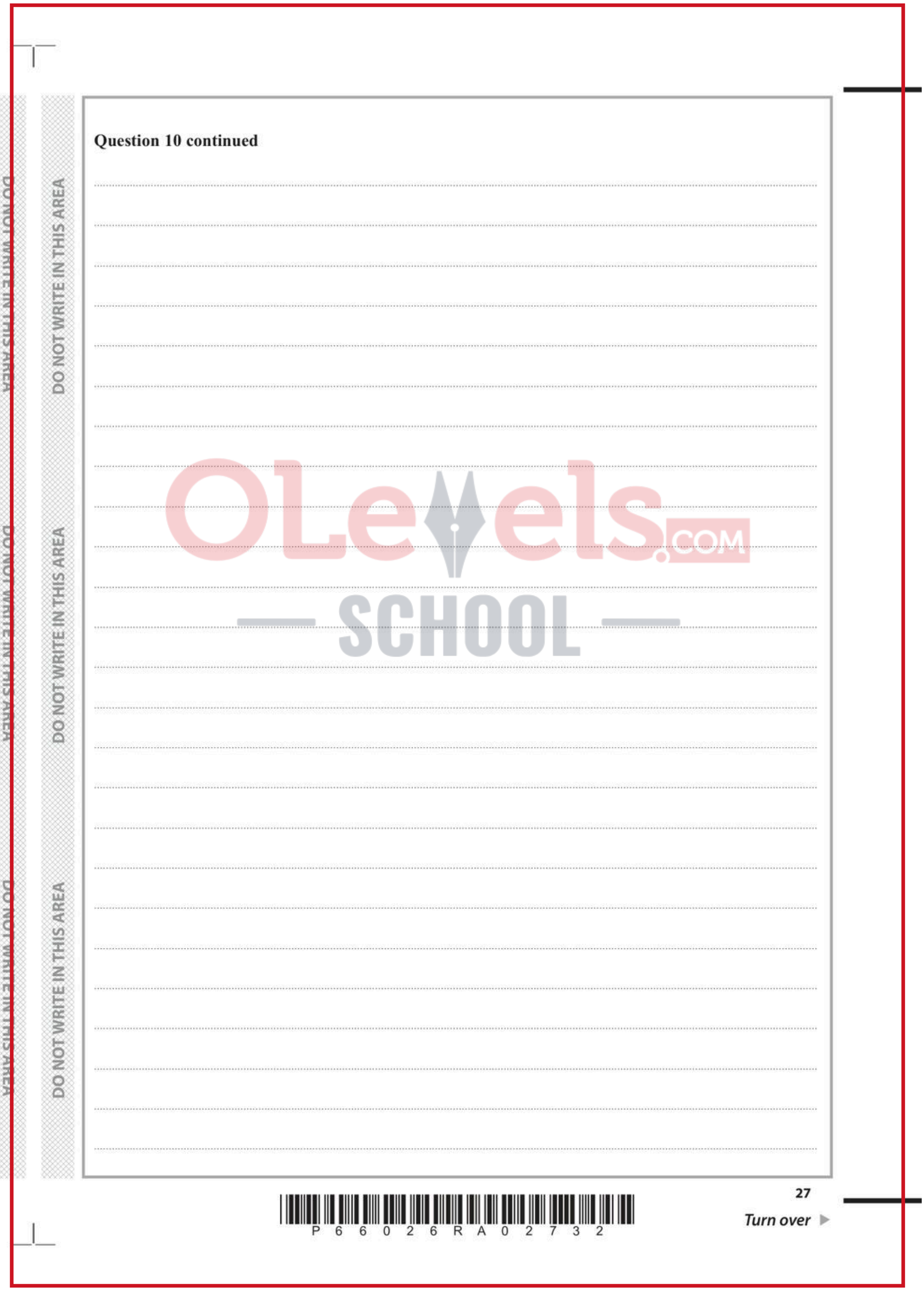
(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.



Question 10 continued

P 6 6 0 2 6 R A 0 2 7 3 2

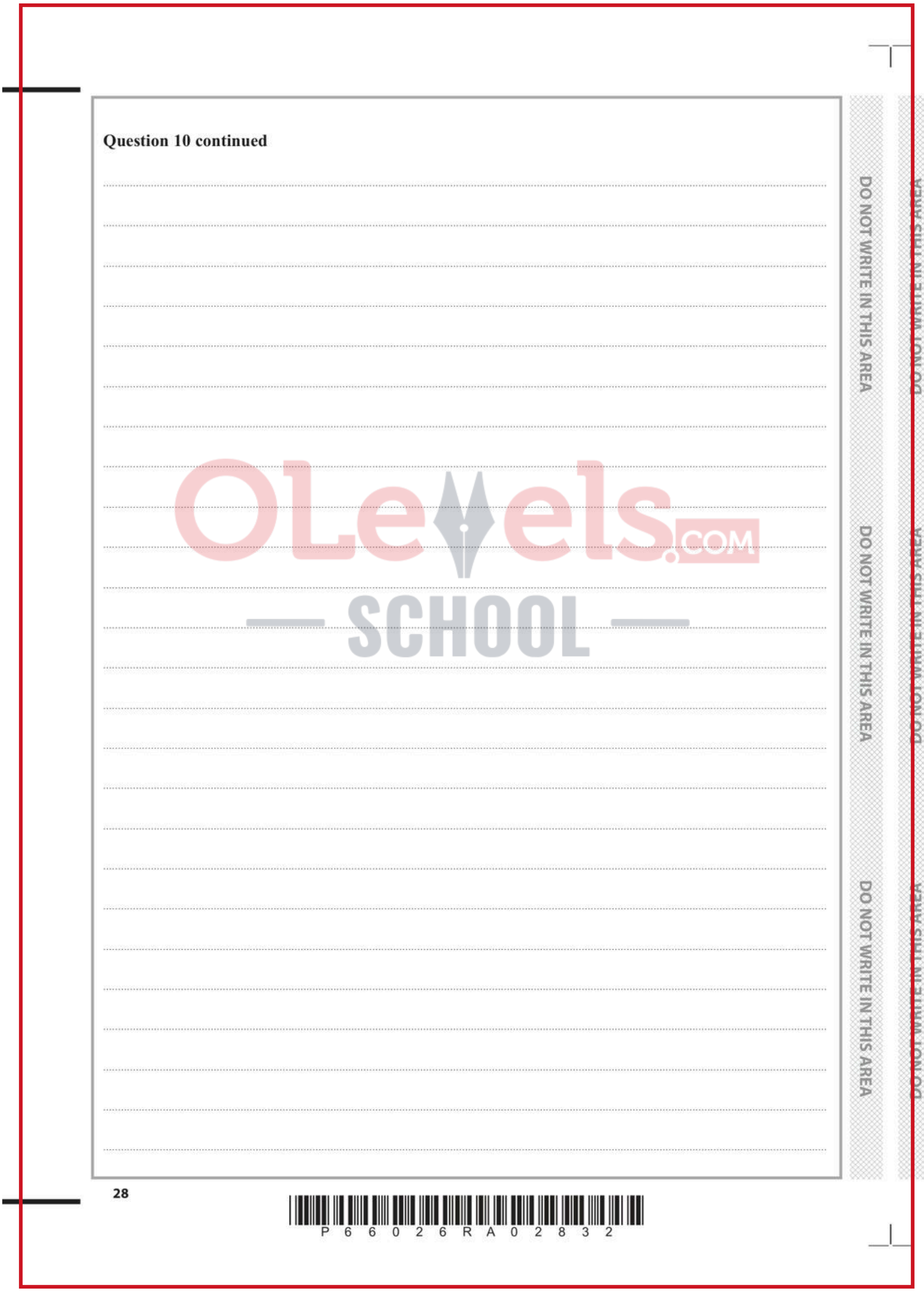
27

Turn over ▶

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.



Question 10 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

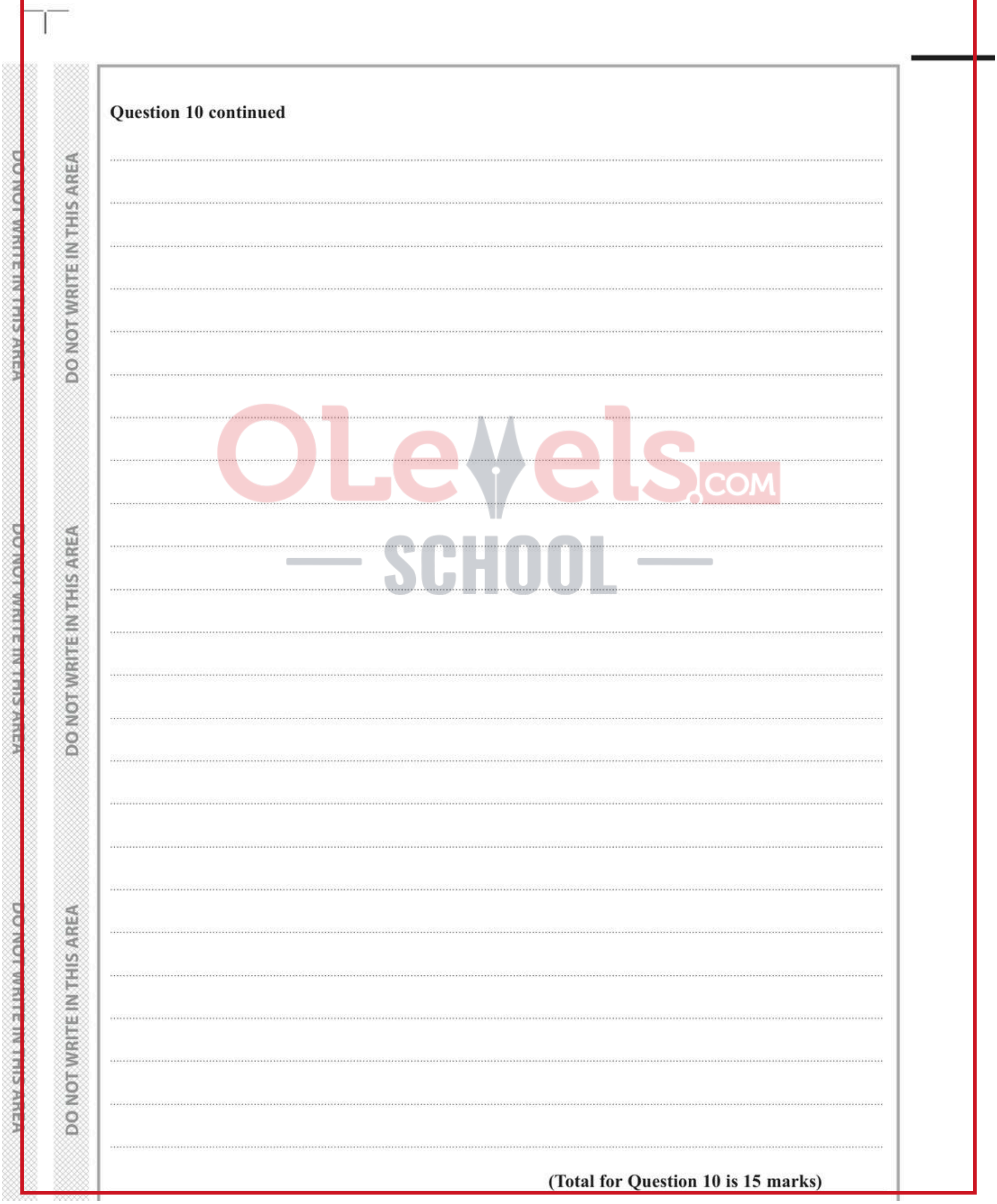
28

P 6 6 0 2 6 R A 0 2 8 3 2

1. Question 10 · part d · 4 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part d only.



Question 10 continued

(Total for Question 10 is 15 marks)

1. Question 10 · part d · 4 marks · Working and answer

Working and answer

2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

9

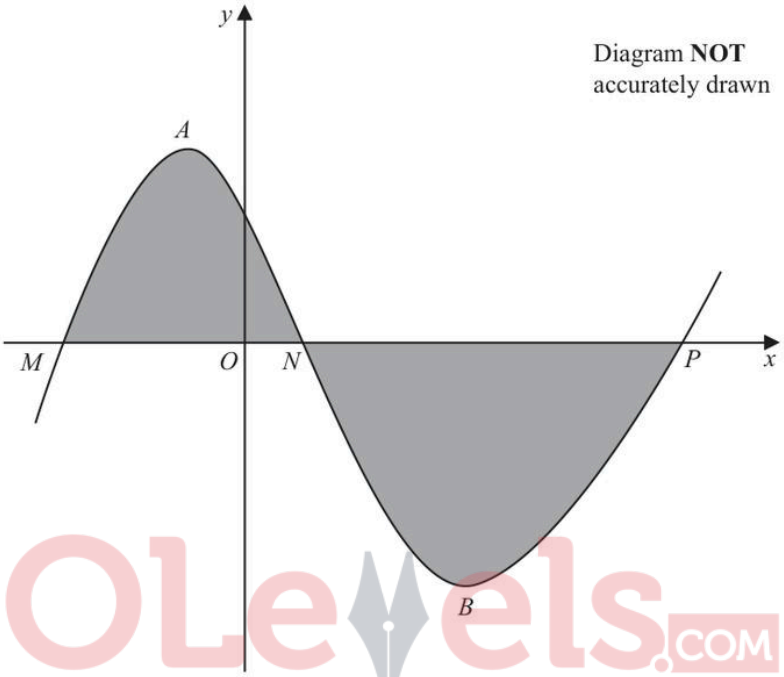


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^3 + ax^2 + bx + c$$

The curve C has a maximum at the point A with coordinates $\left(-\frac{1}{3}, \frac{100}{27}\right)$ and a minimum at the point B with coordinates $(2, -9)$

Given that a, b and c are integers

(a) show that $a = -5$, $b = -4$ and $c = 3$

(5)

(b) (i) Show that $(x+1)$ is a factor of $\mathrm{f}(x)$

(1)

(ii) Hence, or otherwise, use algebra to factorise $\mathrm{f}(x)$ completely.

(3)

The curve C crosses the x-axis at the points M, N and P
The finite regions shown shaded in Figure 4 are bounded by the curve C and parts of the x-axis from M to N and from N to P

(c) Use algebraic integration to determine the total area of the shaded regions.
Give your answer as an exact fraction.

(4)

2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

2. Question 9 · part c · 4 marks

2025 JUN · Paper 01

Further practice · Attempt part c only.

Question 9 continued

(Total for Question 9 is 13 marks)

2. Question 9 · part c · 4 marks · Working and answer

Working and answer

3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

11 $$f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

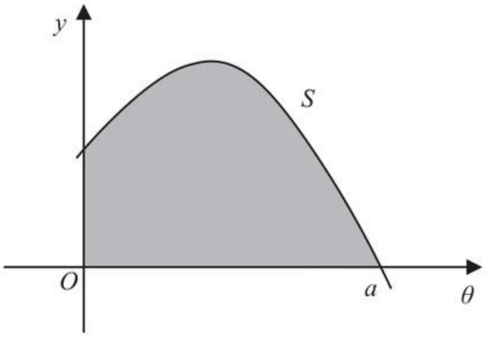


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

3. Question 11 · part c · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 11 continued

(Total for Question 11 is 11 marks)

3. Question 11 · part c · 3 marks · Working and answer

Working and answer

4. Question 3 · part whole · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part whole only.

3 The curve C has equation $y = 9 - x^2$

Use algebraic integration to find the area of the finite region bounded by C and the x-axis.

(5)

4. Question 3 · part whole · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part whole only.

Question 3 continued

(Total for Question 3 is 5 marks)

4. Question 3 · part whole · 5 marks · Working and answer

Working and answer

5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 Using suitable results for $\sin(A+B)$ and $\sin(A-B)$ from the formulae page,

(a) show that $2\sin 4x\cos x = \sin 5x + \sin 3x$

(3)

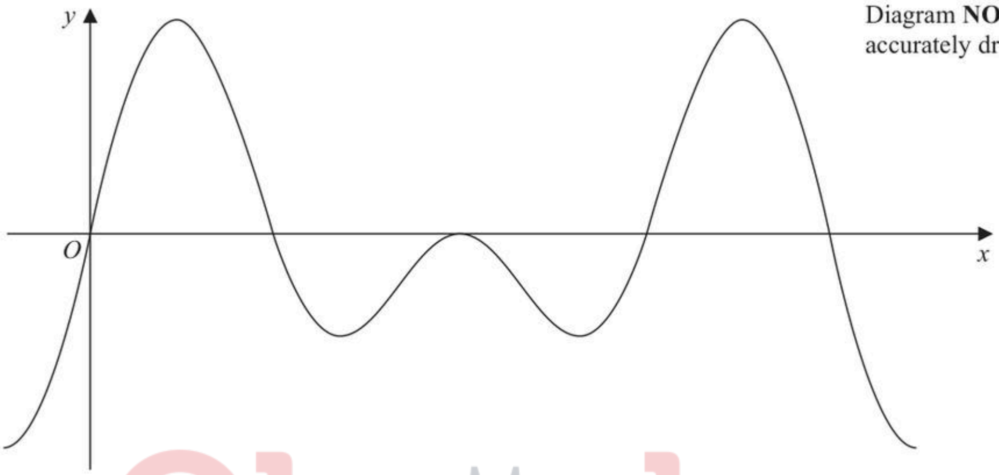


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a sketch of the curve $y = 6\sin 4x\cos x$

(b) Using calculus, find the total area bounded by the curve and the x-axis between

$x = 0$ and $x = \dfrac{\pi}{2}$

Give your answer to 3 significant figures.

(8)

5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

5. Question 10 · part b · 8 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 10 continued

(Total for Question 10 is 11 marks)

5. Question 10 · part b · 8 marks · Working and answer

Working and answer

6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

5 $\quad \mathrm{f}(x) = 2x^3 + ax^2 - 14x + b$ where a and b are constants.

When $\mathrm{f}(x)$ is divided by $(x - 4)$ the remainder is 39

Given that $(x - 1)$ is a factor of $\mathrm{f}(x)$

(a) show that $a = -3$ and find the value of b

(5)

(b) Hence factorise $\mathrm{f}(x)$ completely.

(4)

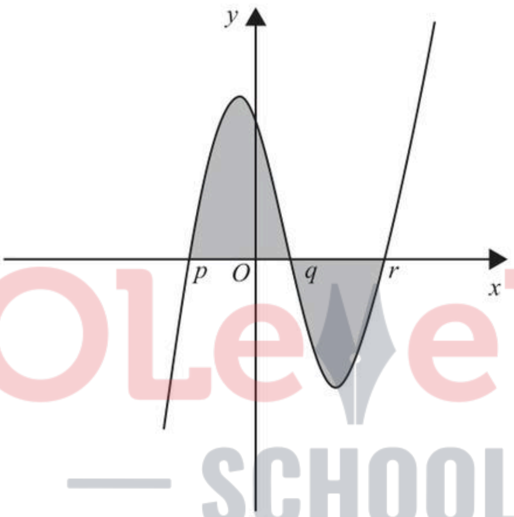


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = \mathrm{f}(x)$

Given that C crosses the x-axis at the points with coordinates $(p, 0)$, $(q, 0)$ and $(r, 0)$

(c) write down the value of p, the value of q and the value of r

(3)

The region shown shaded in Figure 3 is bounded by the curve and the x-axis.

(d) Use algebraic integration to find the exact area of the shaded region.

(4)

6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 5 continued

6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 5 continued

6. Question 5 · part d · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 5 continued

(Total for Question 5 is 16 marks)

6. Question 5 · part d · 4 marks · Working and answer

Working and answer

7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

7 $\text{f}(x) = 2x^3 + Ax^2 + Bx - 20$ where A and B are constants.

Given that

- $(2x+1)$ is a factor of $\text{f}(x)$
- when $\text{f}'(x)$ is divided by $(x-1)$ the remainder is -27

(a) show that $A = 3$ and find the value of B

(5)

(b) Hence, using algebra, solve the equation $\text{f}(x) = 0$

(3)

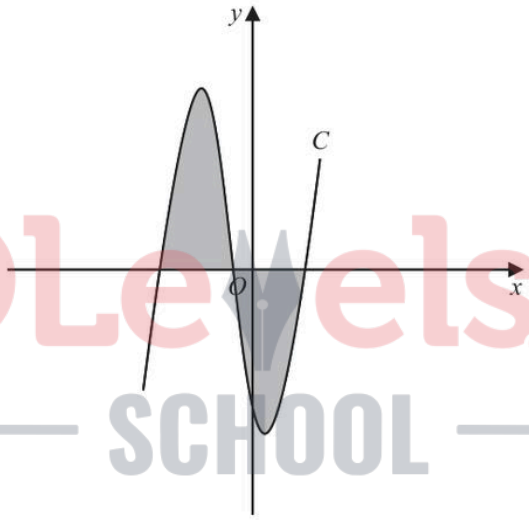


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = \text{f}(x)$

(c) Use calculus to find the exact area of the finite region bounded by C and the x-axis, shown shaded in Figure 2

(4)

7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

7. Question 7 · part c · 4 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 7 continued

(Total for Question 7 is 12 marks)

7. Question 7 · part c · 4 marks · Working and answer

Working and answer

Practice workbook · Area between two curves (or a curve and a line)

Finding the area enclosed by two curves (or a curve and a line) by integrating the top minus the bottom between intersections.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-20

1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

5

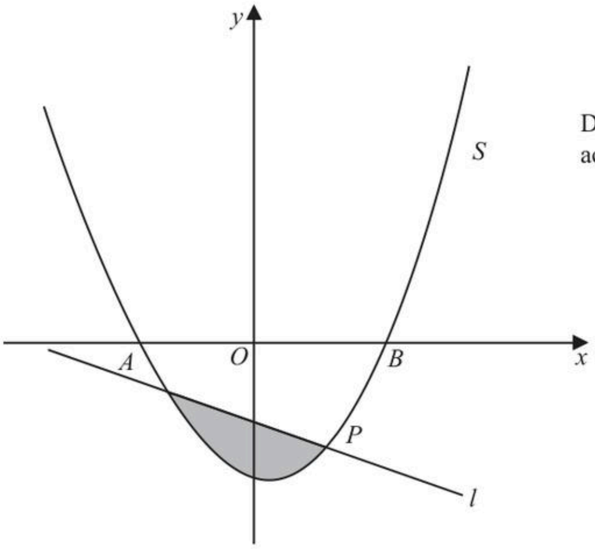


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

1. Question 5 · part c · 7 marks

2023 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 5 continued

(Total for Question 5 is 15 marks)

1. Question 5 · part c · 7 marks · Working and answer

Working and answer

2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

10 $$f(x) = 6x - x^2 \qquad x \in \mathbb{R}$$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

Turn over ▶

2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

2. Question 10 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 10 continued

(Total for Question 10 is 13 marks)

2. Question 10 · part d · 4 marks · Working and answer

Working and answer

3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

7

$$f(x) = x^2 - 9x + 14$$

Given that f(x) can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of f(x)

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 7 continued

Turn over ▶

3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 7 continued

3. Question 7 · part d · 5 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 7 continued

(Total for Question 7 is 13 marks)

3. Question 7 · part d · 5 marks · Working and answer

Working and answer

4. Question 10 · part b · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

10

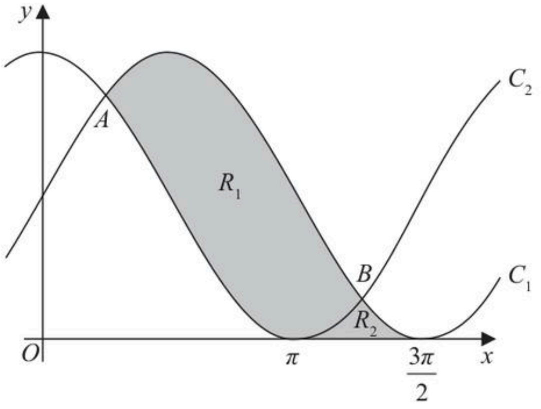


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C_1 with equation $y = \sin x + 1$ and part of the curve C_2 with equation $y = \cos x + 1$

As shown in Figure 2, C_1 and C_2 intersect at the point A and at the point B

(a) Find the exact value of the x coordinate of A and the exact value of the x coordinate of B

(3)

The shaded finite region R_1 shown in Figure 2 is bounded by C_1 and C_2

The shaded finite region R_2 shown in Figure 2 is bounded by the x-axis, C_1 and C_2

(b) Use calculus to find the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

Give your answer in the form $a : \left(\frac{\pi\sqrt{2}}{b} - c\right)$ where a, b and c are integers.

(9)

4. Question 10 · part b · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

Turn over ▶

4. Question 10 · part b · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

4. Question 10 · part b · 9 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

(Total for Question 10 is 12 marks)

4. Question 10 · part b · 9 marks · Working and answer

Working and answer

5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

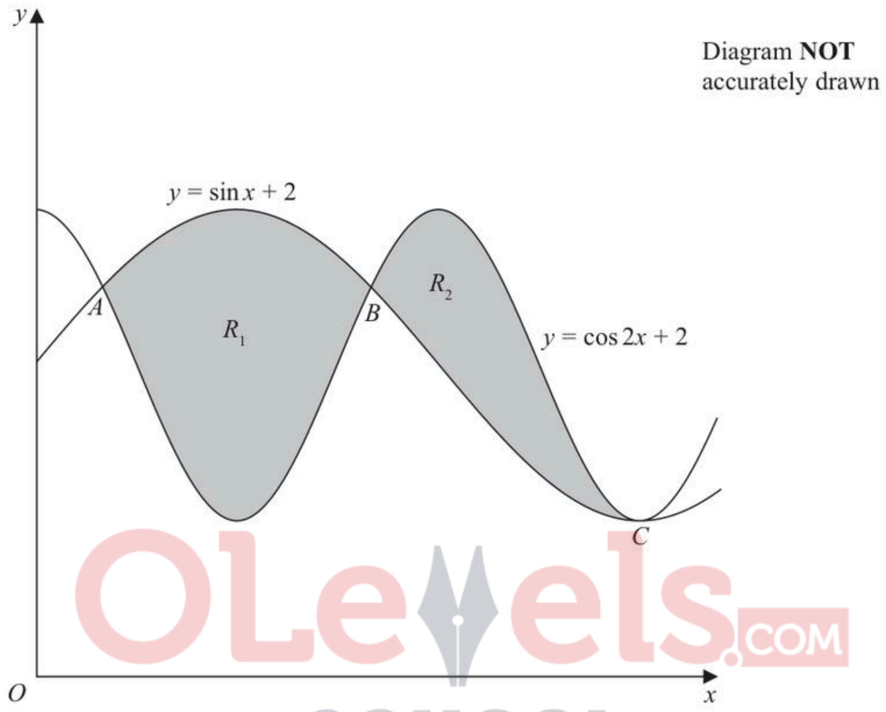


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

$$\text{area of } R_1 : \text{area of } R_2$$

(8)

5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

Question 11 continued

Turn over ▶

5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

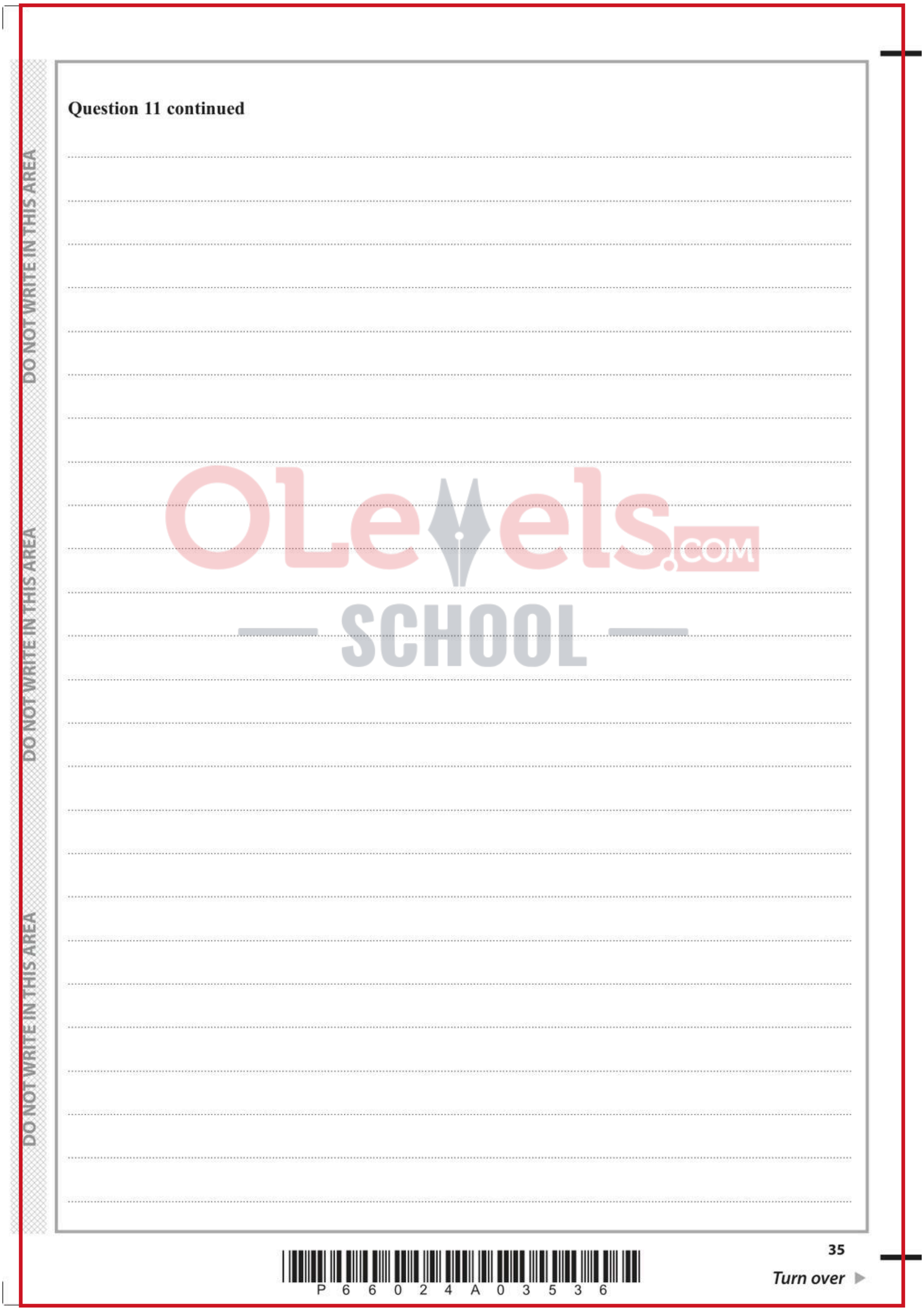
Further practice · Attempt part c only.

Question 11 continued

5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.



Question 11 continued

Turn over ▶

5. Question 11 · part c · 8 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 11 continued

(Total for Question 11 is 15 marks)

5. Question 11 · part c · 8 marks · Working and answer

Working and answer

6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

9

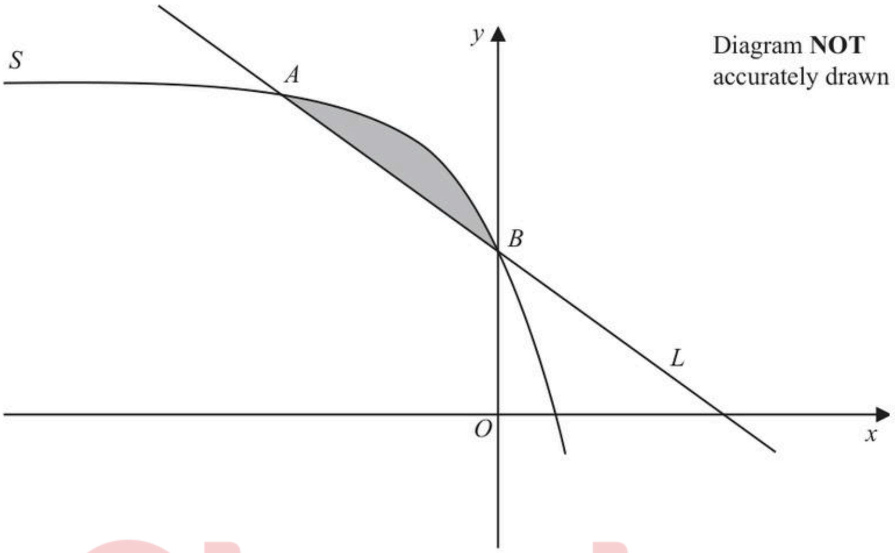


Figure 3

Figure 3 shows a sketch of part of the curve S with equation $y = -2e^{3x} + 4$ and the line L

The curve S has intersections with the line L at the points A and B with x coordinates $x = -1$ and $x = 0$ respectively.

The finite region bounded by S and L is shown shaded in Figure 3

Use calculus to find the exact area of this region.

Give your answer in the form $\dfrac{a + be^{-c}}{c}$ where a, b and c are integers to be found.

(8)

6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

6. Question 9 · part whole · 8 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 9 continued

(Total for Question 9 is 8 marks)

6. Question 9 · part whole · 8 marks · Working and answer

Working and answer

7. Question 9 · part b · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

9 $$f(x) = 3x^4 + 4x^3 - 36x^2 + 64$$

Given that $f(x)$ can be written in the form $(x - 2)^2(ax^2 + bx + c)$

(a) find the value of a, the value of b and the value of c

(4)

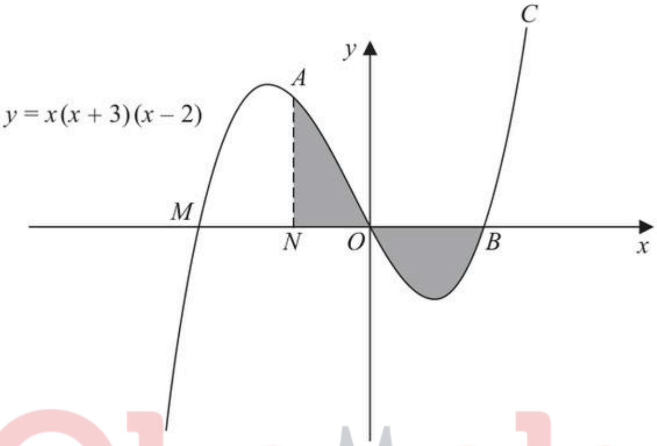


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = x(x + 3)(x - 2)$

The curve C crosses the x-axis at the point M, the origin and the point B.
The point N lies on the x-axis between M and O.
The point A lies on C such that AN is parallel to the y-axis.

The area of the shaded region bounded by the curve and OB is numerically equal to the area of the shaded region bounded by the curve, ON and NA.

Given that the coordinates of N are $(n, 0)$,

(b) use algebraic integration to show that n satisfies the equation

$$(x - 2)^2(3x^2 + 16x + 16) = 0$$

(7)

(c) Hence find the exact coordinates of A.

(5)

7. Question 9 · part b · 7 marks · Working and answer

Working and answer

8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

11 $$f(x) = 10 + 6x - x^2$$

Given that $f(x)$ can be written in the form $A(x+B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

Question 11 continued

8. Question 11 · part d · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part d only.

Question 11 continued

(Total for Question 11 is 14 marks)

8. Question 11 · part d · 5 marks · Working and answer

Working and answer

9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 The equation of a curve C is $y = \mathrm{f}(x)$ where $\mathrm{f}'(x) = 3x^2 - 4x - p$ and $p \neq 0$

The points with coordinates $(2, 0)$ and $(-1, 9)$ lie on C.

(a) Show that C has equation $y = x^3 - 2x^2 - 4x + 8$

(6)

The straight line l has equation $y = 8 - 4x$

(b) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(6)

9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

P 6 0 4 7 6 A 0 3 3 4 0

Turn over ▶

9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

9. Question 10 · part b · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 10 continued

(Total for Question 10 is 12 marks)

9. Question 10 · part b · 6 marks · Working and answer

Working and answer

10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

8

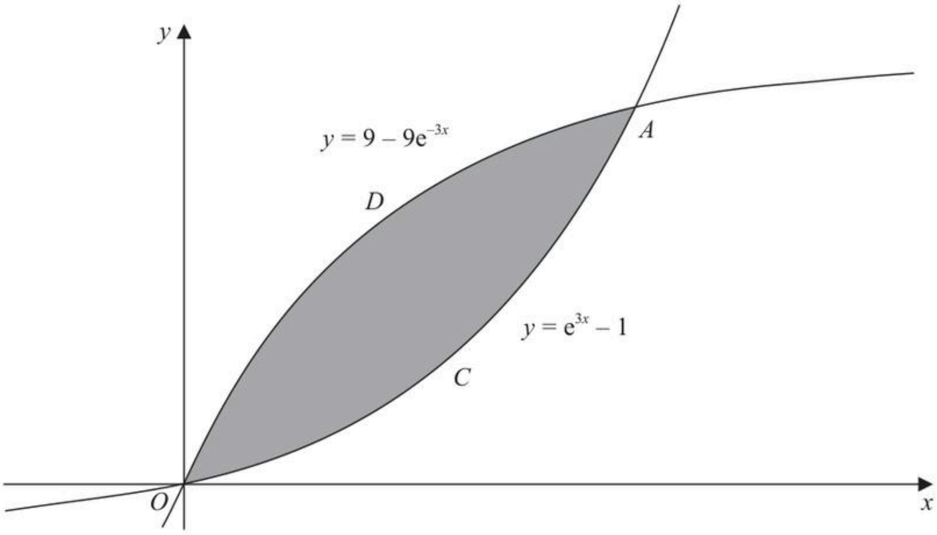


Figure 2

Figure 2 shows part of the curve C with equation $y = e^{3x} - 1$ and part of the curve D with equation $y = 9 - 9e^{-3x}$

The curves intersect at the origin O and the point A.

(a) (i) Show that the x coordinate of the point A satisfies the equation

$$(e^{3x})^2 - 10e^{3x} + 9 = 0$$

(ii) Hence, show that the x coordinate of the point A is $\frac{1}{3}\ln 9$

(5)

The finite region bounded by C and by D is shown shaded in Figure 2.

(b) Use calculus to find the exact area of this region.

(6)

10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

Turn over ▶

10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

10. Question 8 · part b · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 8 continued

(Total for Question 8 is 11 marks)

10. Question 8 · part b · 6 marks · Working and answer

Working and answer

11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

8

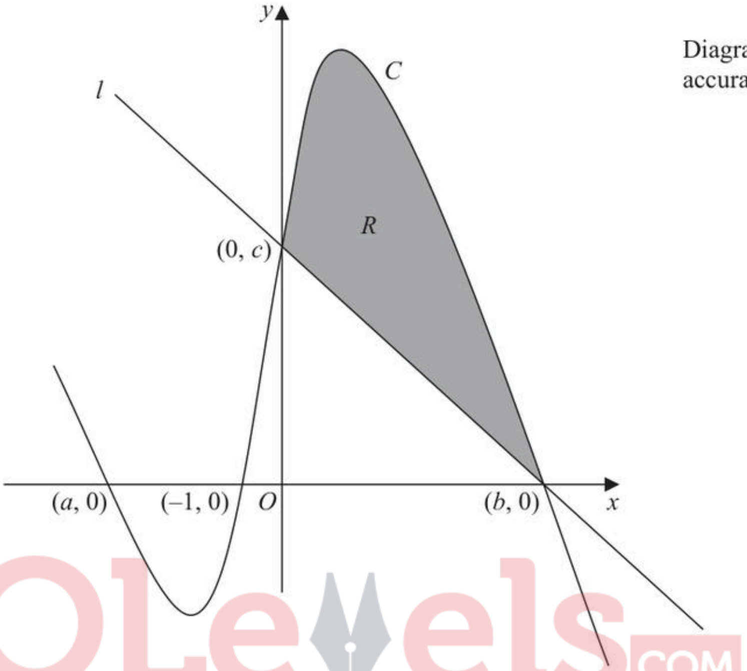


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = \mathrm{f}(x)$

The curve C passes through the points with coordinates

$$(a, 0), (-1, 0), (b, 0) \text{ and } (0, c)$$

Given that $\mathrm{f}'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

11. Question 8 · part c · 5 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 8 continued

(Total for Question 8 is 15 marks)

11. Question 8 · part c · 5 marks · Working and answer

Working and answer

Practice workbook · Volumes of revolution between two boundaries

Finding volumes about the x-axis for regions between a curve and a line or two curves, by subtracting or adding solids.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: ______________________ Class: ___________ Date: ___________

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

Type ID: FPM-CAL-23

1. Question 2 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

2 The finite region enclosed by the curve with equation $y = 4 - x^2$ and the line with equation $y = x + 2$ is rotated through 360° about the x-axis.

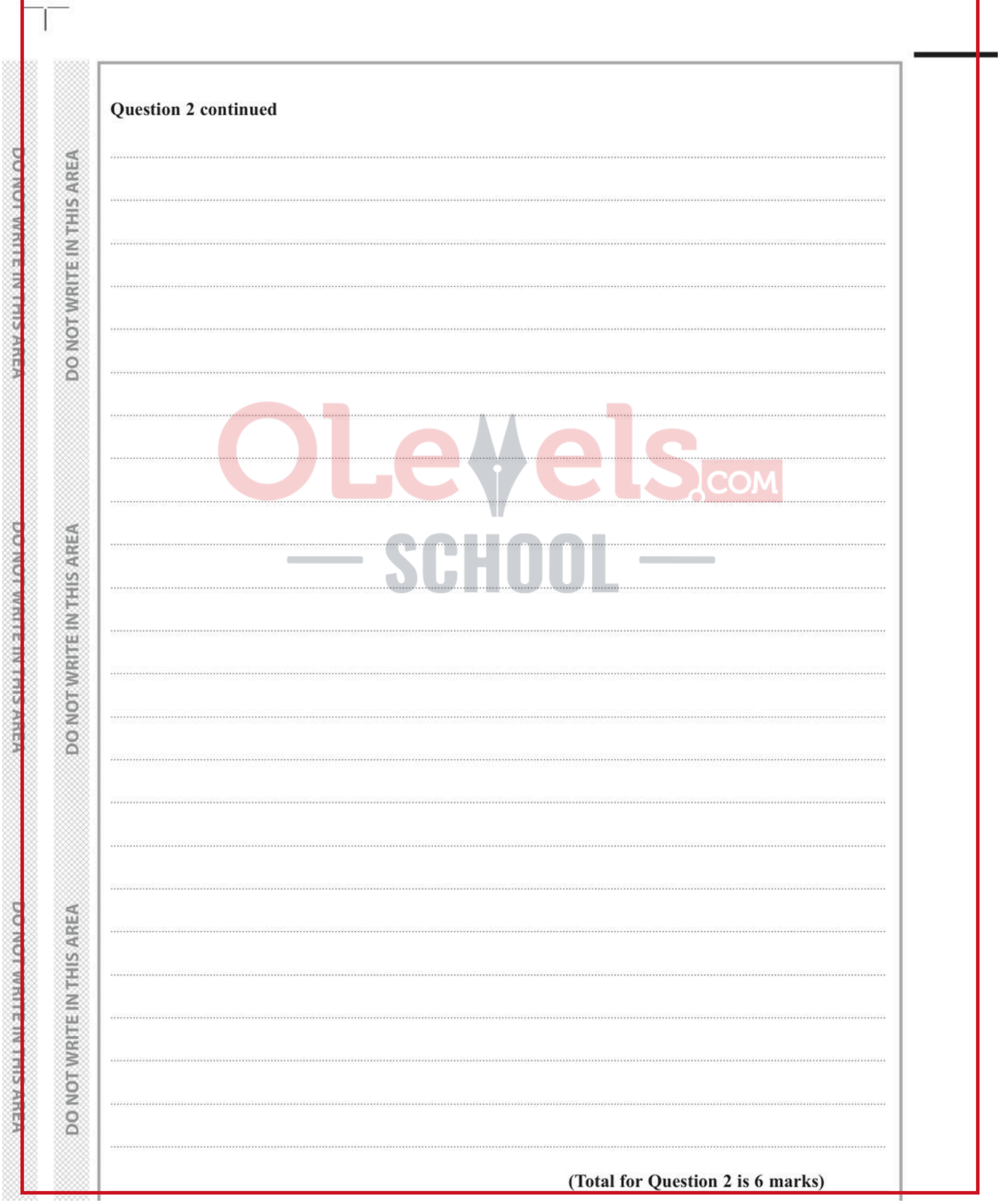
Use algebraic integration to find the exact volume of the solid formed.

(6)

1. Question 2 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.



Question 2 continued

(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

10

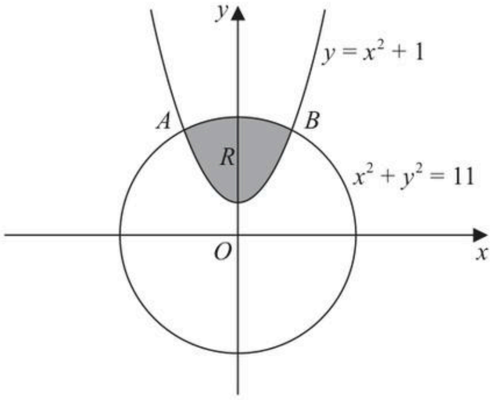


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = x^2 + 1$ and the curve with equation $x^2 + y^2 = 11$

The two curves intersect at the point A and at the point B.

(a) Find the x coordinate of the point A and the x coordinate of the point B.

(4)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume, to 2 decimal places, of the solid generated.

(5)

2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 10 continued

2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 10 continued

2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 10 continued

(Total for Question 10 is 9 marks)

2. Question 10 · part b · 5 marks · Working and answer

Working and answer

3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

10 $$f(x) = 32x^3 - 33x + 1$$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

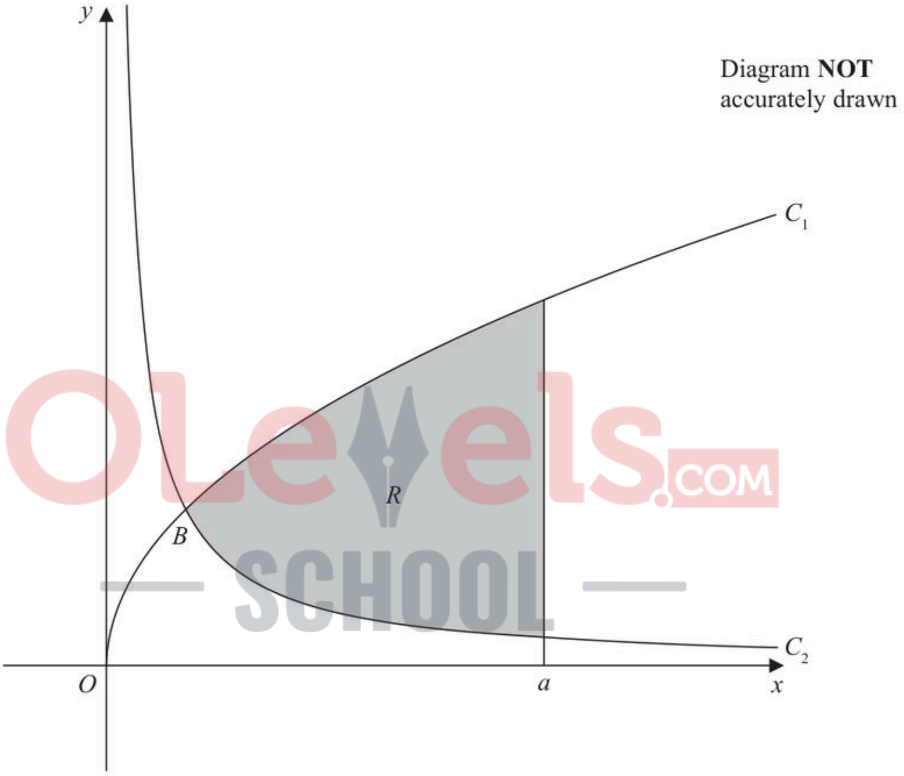


Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

Turn over ▶

3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

Turn over ▶

3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

(Total for Question 10 is 14 marks)

3. Question 10 · part d · 7 marks · Working and answer

Working and answer

4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

11

Diagram **NOT** accurately drawn

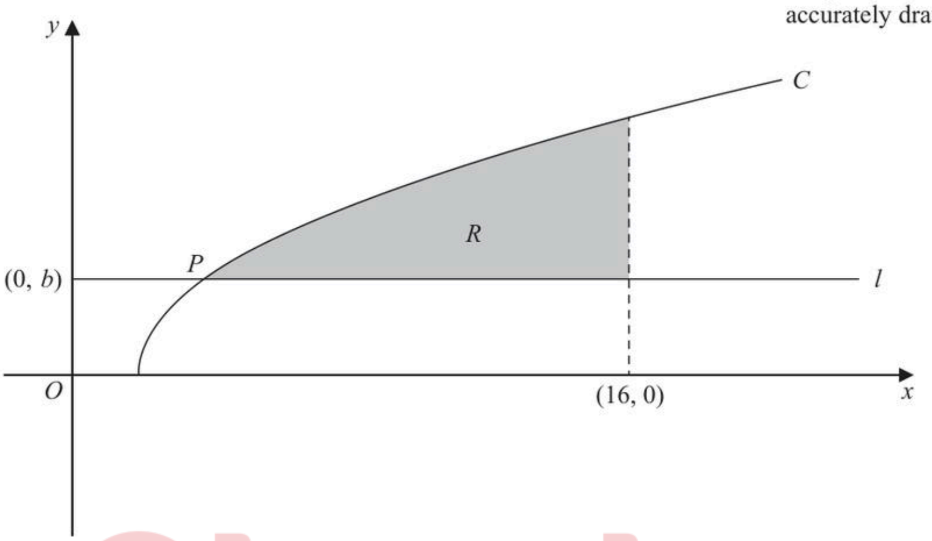


Figure 2

Figure 2 shows part of the curve C with equation $y = \sqrt{x-2}$
Figure 2 also shows the straight line l with equation $y = b$ for $x > 0$ where $b > 0$

Given that C and l intersect at the point P with coordinates (a, b), where $2 < a < 16$

(a) show that $b^2 = a - 2$

(2)

The finite region R bounded by C, the straight line with equation $x = 16$ and l, shown shaded in Figure 2, is rotated through 360° about the x-axis to form a solid S.

Given that the volume of the solid formed is 50π

(b) use algebraic integration to find the value of a and the value of b.

(9)

4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

Turn over ▶

4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

Turn over ▶

4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

(Total for Question 11 is 11 marks)

4. Question 11 · part b · 9 marks · Working and answer

Working and answer

5. Question 11 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

11

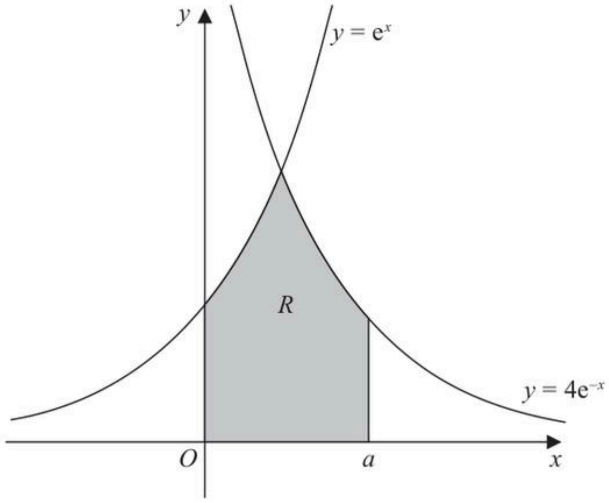


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve with equation $y = e^x$, the curve with equation $y = 4e^{-x}$, the straight line with equation $x = a$, the x-axis and the y-axis.

When the region R is rotated through 360° about the x-axis, the volume of the solid generated is

$$k - 8\pi e^{-4}$$

where k is a constant.

Using algebraic integration, find a possible value of a and the exact corresponding value of k.

(8)

5. Question 11 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

Question 11 continued

5. Question 11 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

Question 11 continued

(Total for Question 11 is 8 marks)

5. Question 11 · part whole · 8 marks · Working and answer

Working and answer

6. Question 10 · part b · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

10

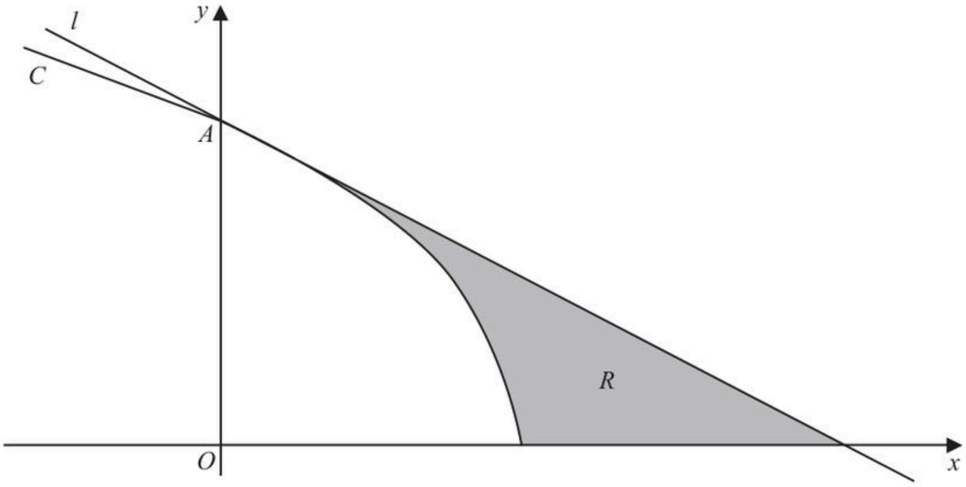


Figure 3

Figure 3 shows part of the curve C with equation $y = 3\sqrt{4-x}$ and part of the line l with equation $px + qy + r = 0$ where p and r are integers and q is prime.

The curve C cuts the y-axis at the point A

The line l is the tangent to C at A

(a) Find the value of p, the value of q and the value of r

(5)

The finite region R, shown shaded in Figure 3, is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume of the solid formed.
Give your answer in terms of π

(5)

6. Question 10 · part b · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

6. Question 10 · part b · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

6. Question 10 · part b · 5 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

Question 10 continued

(Total for Question 10 is 10 marks)

6. Question 10 · part b · 5 marks · Working and answer

Working and answer

7. Question 9 · part whole · 10 marks

2024 NOV · Paper 01

Further practice · Attempt part whole only.

9

Diagram **NOT** accurately drawn

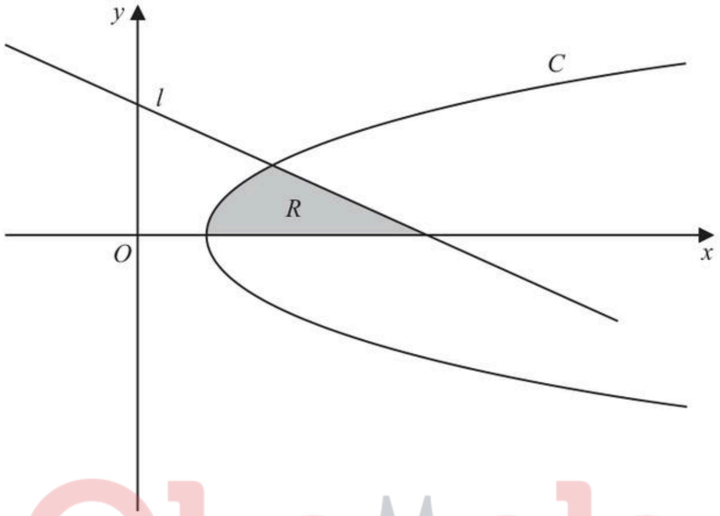


Figure 3

Figure 3 shows part of the curve C with equation $y^2 = x - 1$ and part of the line l with equation $2y + x - 4 = 0$

The region R, bounded by the x-axis, the curve C and the line l, is rotated through 360° about the x-axis.

Using algebraic integration, find the exact value of the volume of the solid generated.

(10)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

7. Question 9 · part whole · 10 marks

2024 NOV · Paper 01

Further practice · Attempt part whole only.

Question 9 continued

7. Question 9 · part whole · 10 marks

2024 NOV · Paper 01

Further practice · Attempt part whole only.

Question 9 continued

7. Question 9 · part whole · 10 marks

2024 NOV · Paper 01

Further practice · Attempt part whole only.

Question 9 continued

(Total for Question 9 is 10 marks)

7. Question 9 · part whole · 10 marks · Working and answer

Working and answer

8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

6

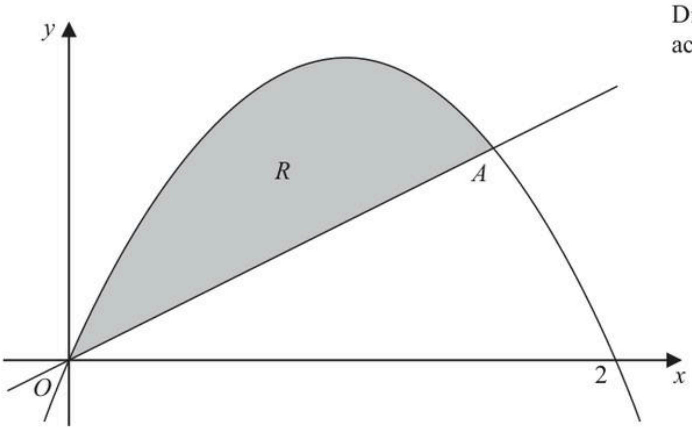


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = 2x - x^2$ and the line with equation $2y - x = 0$

The curve and the line intersect at the origin O and the point A.

(a) Show that the point A has coordinates $\left(\frac{3}{2}, \frac{3}{4}\right)$.

(2)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find, in terms of π, the volume of the solid formed.

(6)

8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

Turn over ▶

8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

(Total for Question 6 is 8 marks)

8. Question 6 · part b · 6 marks · Working and answer

Working and answer

9. Question 5 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

5

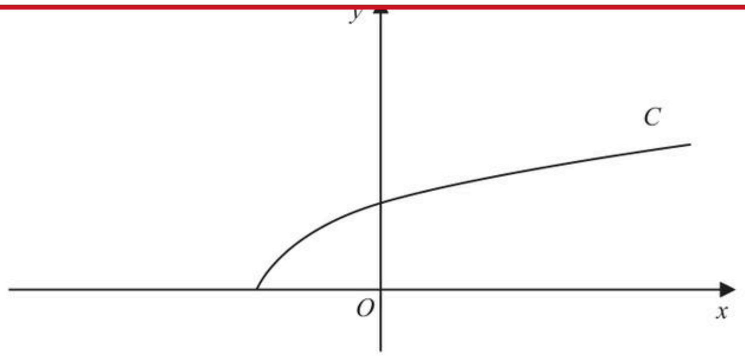


Figure 2

Figure 2 shows the graph of part of the curve C with equation $y=\sqrt{2x+6}$
The finite region enclosed by the curve C and the straight line with equation $3y-x=3$ is rotated through 360° about the x-axis.

Use algebraic integration to find the exact volume of the solid generated.
Give your answer in terms of π

(8)

9. Question 5 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

Question 5 continued

(Total for Question 5 is 8 marks)

9. Question 5 · part whole · 8 marks · Working and answer

Working and answer

10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \dfrac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \dfrac{\cos 2A+1}{2}$

(5)

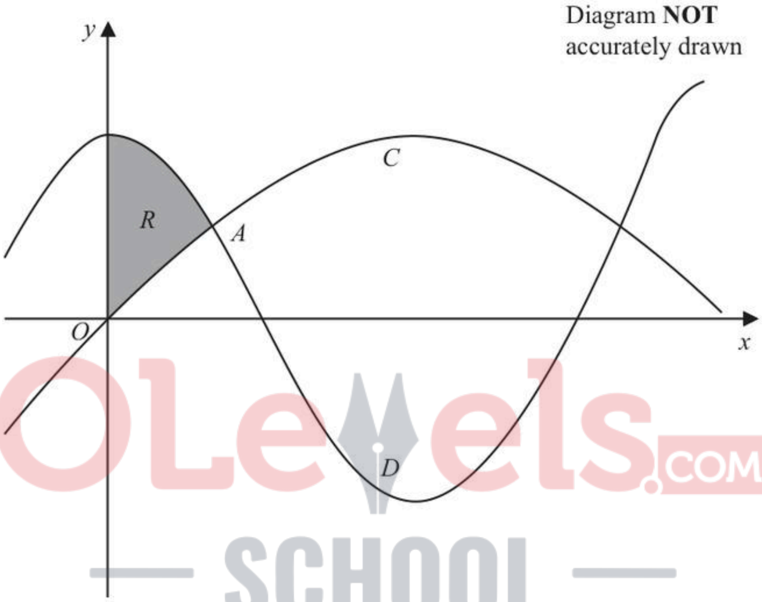


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$

Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through 360° about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\dfrac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

(Total for Question 10 is 15 marks)

10. Question 10 · part c · 5 marks · Working and answer

Working and answer